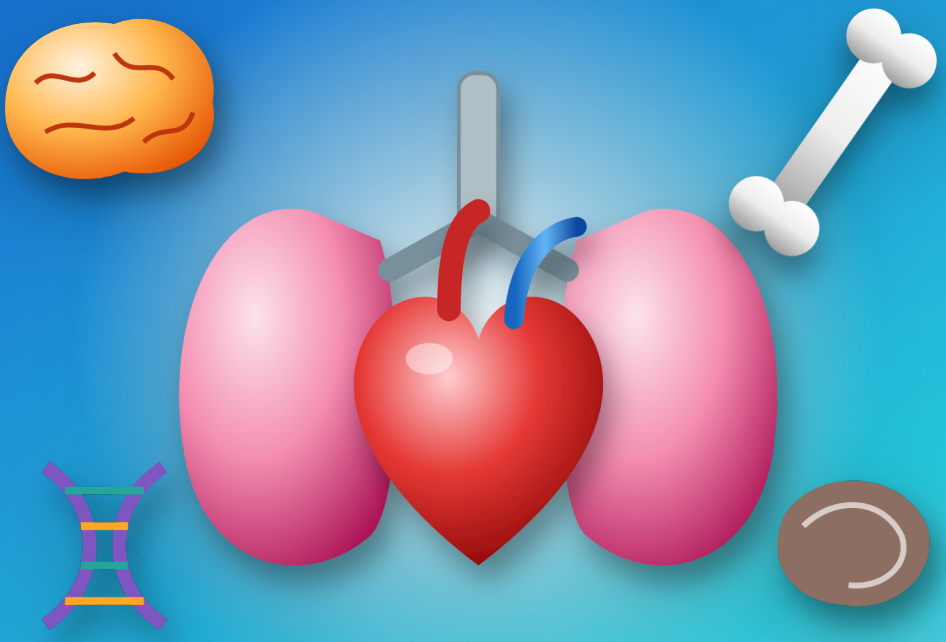
GSSSB · STAFF NURSE EXAM

# Anatomy & Physiology for Nurses

Complete Colourful Notes · 3D Diagrams · Mnemonics · Exam Points



Kidney

13 Chapters | 3D Coloured Diagrams | All body systems | Normal values · Physiology | Clinical correlations | Exam Points (PYQ style)

Hinglish explanation + English exam terms



**Sister Asha**
Aapki Nursing teacher

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★ **Key Points** – sabse important facts

**Asha Didi ki Tip** – samjhane wali baat

**! Nursing Alert** – galti mat karna

✎ **Exam Point** – PYQ style, baar baar aata hai

♫ **Mnemonic** – yaad karne ki trick

**D Definition** – exact definition

# 1 Introduction, Cell & Tissues

## 1.1 Basic Terminology

**D Definitions**

- **Anatomy** – study of **structure** of the body (Greek **ana** = up, **tome** = cutting); **Father of Anatomy – Andreas Vesalius** (**De Humani Corporis Fabrica**, 1543); **Father of Medicine – Hippocrates**.
- **Physiology** – study of **functions** of the body; **Father of (experimental) Physiology – Claude Bernard** ("milieu intérieur"); **William Harvey – discovered blood circulation (1628)**; **Walter Cannon – term "homeostasis"**.
- **Homeostasis** – maintenance of a **constant internal environment**; mostly by **negative feedback** (temperature, BP, blood glucose); **positive feedback** examples – **labour contractions (oxytocin), blood clotting, LH surge, milk ejection**.

- **Levels of organisation** – chemical → **cell (structural & functional unit of life)** → tissue → organ → system → organism.
- **Anatomical position** – body **erect, face forward, arms by side, palms facing forward**, feet together.
- **Directional terms** – **superior/ cranial** (towards head), **inferior/ caudal**, **anterior/ ventral**, **posterior/ dorsal**, **medial** (towards midline), **lateral**, **proximal** (near origin/ trunk), **distal**, **superficial**, **deep**, **ipsilateral/ contralateral**.

| Plane | Divides body into |
|---|---|
| **Sagittal (median/ midsagittal)** | **Right & left** (equal halves if midsagittal) |
| **Frontal (coronal)** | **Anterior & posterior** |
| **Transverse (horizontal/ axial)** | **Superior & inferior** |
| Oblique | At an angle |

### Body cavities

- **Dorsal cavity** – **cranial (brain)** + **vertebral/ spinal canal (spinal cord)**.
- **Ventral cavity** – **thoracic** (2 pleural + **mediastinum** – heart in **pericardial** cavity, trachea, oesophagus, thymus, great vessels) separated by **diaphragm** from **abdominopelvic** cavity.
- **Abdomen – 9 regions** – **right hypochondrium** (liver, gall bladder), **epigastrium** (stomach, pancreas), **left hypochondrium** (spleen), **right lumbar, umbilical, left lumbar**, **right iliac (appendix – McBurney's point)**, **hypogastrium/ suprapubic** (bladder, uterus), **left iliac** (sigmoid colon); **4 quadrants** – RUQ, LUQ, RLQ, LLQ.

## 1.2 The Cell

- **Cell theory – Schleiden & Schwann (1838–39)**; **"cell" word – Robert Hooke (1665)**; human body ≈ **37 trillion cells**.
- **Largest cell – ovum**; **smallest – sperm (male gamete)/ RBC among smallest**; **longest – neuron**; cells without nucleus – **mature RBC, platelets**; multinucleated – **skeletal muscle, osteoclasts**.

| Organelle | Function |
|---|---|
| **Cell (plasma) membrane** | **Fluid mosaic model (Singer & Nicolson)** – phospholipid bilayer + proteins; **selectively permeable** |
| **Nucleus** | **Control centre**; contains DNA (chromatin), **nucleolus** – makes **ribosomes (rRNA)** |
| **Mitochondria** | **"Powerhouse" – ATP by aerobic respiration (Krebs cycle, ETC)**; have own DNA (**maternal inheritance**) |
| **Ribosomes** | **Protein synthesis** (free or on rough ER) |
| **Rough ER** | Protein synthesis & transport |
| **Smooth ER** | **Lipid & steroid synthesis, detoxification (liver), $Ca^{2+}$ storage (sarcoplasmic reticulum in muscle)** |
| **Golgi apparatus** | **Packaging, modification & secretion ("post office")** |
| **Lysosomes** | **"Suicide bags" – digestive enzymes**, autophagy |
| **Peroxisomes** | Break down $H_2O_2$, fatty acids |
| **Centrosome/ centrioles** | **Spindle formation in cell division** (absent in mature neurons) |
| **Cilia/ flagella** | Movement (respiratory tract cilia, fallopian tube; sperm flagellum) |
| **Cytoskeleton** | Microtubules, microfilaments – shape & movement |

## Transport across membrane

- **Passive (no ATP)** – **diffusion** (high → low concentration; $O_2$, $CO_2$), **facilitated diffusion** (carrier – **glucose**), **osmosis** (**water** from low solute → high solute concentration across semipermeable membrane), **filtration** (pressure – **glomerulus, capillaries**).
- **Active (ATP)** – **$Na^+$-$K^+$ pump (3 $Na^+$ out, 2 $K^+$ in)**, **endocytosis** (phagocytosis – cell eating, pinocytosis – cell drinking), **exocytosis**.
- **Tonicity** – **isotonic (0.9 % NaCl, 5 % dextrose initially)** – no change; **hypotonic** – cell **swells/ bursts (haemolysis)**; **hypertonic** – cell **shrinks (crenation)**.
- **Main intracellular ion – $K^+$ (and phosphate)**; **main extracellular ion – $Na^+$ (and $Cl^-$)**.

## Cell division & genetics

- **Mitosis** – somatic cells; **2 identical diploid (46) daughter cells**; phases **PMAT – prophase, metaphase (chromosomes best seen – karyotyping), anaphase, telophase**; interphase (G1, S – DNA replication, G2).
- **Meiosis** – gametes; **4 haploid (23) cells**; **crossing over** (prophase I) → variation.
- **Chromosomes – 46 (23 pairs: 22 autosomes + 1 sex)**; **male XY, female XX**; **DNA double helix – Watson & Crick (1953)**; bases **A=T (2 H-bonds), G≡C (3)**; RNA has **uracil** instead of thymine.
- **Gene** – unit of heredity; **genotype vs phenotype**; **dominant/ recessive**; **Down syndrome – trisomy 21**, **Turner – 45 XO**, **Klinefelter – 47 XXY**.

## 1.3 Tissues

| Tissue | Types / examples |
|---|---|
| **Epithelial** (covers, lines; avascular) | **Simple squamous** (alveoli, endothelium, Bowman's capsule), **simple cuboidal** (kidney tubules, thyroid follicles), **simple columnar** (stomach, intestine – with microvilli), **ciliated columnar** (fallopian tube), **pseudostratified ciliated columnar** (trachea – respiratory epithelium), **stratified squamous keratinised (skin)**, **non-keratinised (mouth, oesophagus, vagina, cornea)**, **transitional (urinary bladder, ureter – stretch)** |
| **Connective** (most abundant, supports) | **Loose (areolar), adipose**, dense (tendons, ligaments), **cartilage** (hyaline – most common: costal, trachea, articular; **elastic – ear pinna, epiglottis**; **fibrocartilage – intervertebral disc, pubic symphysis, menisci**), **bone**, **blood (fluid connective tissue)**, lymph |
| **Muscular** | **Skeletal** (voluntary, striated), **cardiac** (involuntary, striated, intercalated discs), **smooth** (involuntary, non-striated – viscera) |
| **Nervous** | Neurons + neuroglia |

- **Membranes** – **mucous** (lines openings to exterior), **serous** (**pleura, pericardium, peritoneum** – 2 layers, parietal & visceral), **synovial** (joints), **cutaneous (skin)**.
- **Glands** – **exocrine** (ducts – sweat, salivary), **endocrine** (ductless – hormones into blood), **mixed – pancreas, gonads**; **merocrine/ apocrine/ holocrine (sebaceous)**.
- **Regeneration** – epithelium, liver, bone, blood regenerate well; **neurons & cardiac muscle – poor**.

**Yaad karne ki trick**

- Mitosis phases – **"PMAT"** (**P**lease **M**ake **A** **T**ea).
- Positive feedback – **"COLB"** – **C**lotting, **O**xytocin (labour), **L**H surge, **B**reast milk ejection.
- Transitional epithelium – **"T for Toilet"** – urinary bladder.

**PYQ style**

- Structural & functional unit of life – **cell**; powerhouse – **mitochondria**; suicide bags – **lysosomes**.
- Protein synthesis – **ribosomes**; packaging – **Golgi**; detoxification – **smooth ER**.
- Father of anatomy – **Vesalius**; blood circulation discovered – **William Harvey**.
- Homeostasis term – **Walter Cannon**; positive feedback – **labour contractions**.
- Plane dividing body into right & left – **sagittal**; anterior & posterior – **frontal (coronal)**.
- Appendix region – **right iliac fossa**; spleen – **left hypochondrium**.
- Largest cell – **ovum**; longest – **neuron**; no nucleus – **mature RBC**.
- Chromosomes in human – **46**; gametes – **23 (haploid)**.
- Water movement across semipermeable membrane – **osmosis**.
- $Na^+$-$K^+$ pump – **3 $Na^+$ out, 2 $K^+$ in (active)**.
- Main intracellular cation – **$K^+$**; extracellular – **$Na^+$**.
- Urinary bladder epithelium – **transitional**; trachea – **pseudostratified ciliated columnar**.
- Ear pinna cartilage – **elastic**; intervertebral disc – **fibrocartilage**.
- Fluid connective tissue – **blood**.
- Down syndrome – **trisomy 21**.

# 2 Skeletal System & Joints

## 2.1 Functions & Bone Basics

- **Functions** – support, **protection** (skull – brain, ribs – heart & lungs), movement (levers), **haemopoiesis (red marrow)**, **mineral storage (99 % calcium, 85 % phosphorus)**, fat storage (yellow marrow), endocrine (osteocalcin).
- **Adult bones – 206** (axial **80** + appendicular **126**); **newborn ≈ 270–300** (fusion later).
- **Largest/ longest & strongest bone – femur**; **smallest – stapes (ear ossicle)**; **longest bone of upper limb – humerus**; **only bone with no articulation with other bone – hyoid**; **most commonly fractured bone – clavicle** (also commonest in birth injury); **first bone to ossify – clavicle**; **hardest substance – tooth enamel**.

| Type | Example |
|---|---|
| **Long** | Femur, humerus, tibia, radius, phalanges |
| **Short** | **Carpals, tarsals** |
| **Flat** | Skull bones, **sternum, ribs, scapula** |
| **Irregular** | **Vertebrae**, hip bone, some facial bones |
| **Sesamoid** (in tendon) | **Patella (largest sesamoid)**, pisiform |
| **Pneumatic** (air sinuses) | Maxilla, frontal, ethmoid, sphenoid |

### Bone structure

- **Long bone** – **diaphysis (shaft)** with **medullary cavity**, **epiphysis** (ends), **metaphysis**, **epiphyseal (growth) plate – longitudinal growth**, **periosteum** (outer – nutrition, growth in thickness, repair; **sensitive to pain**), endosteum, articular (hyaline) cartilage.
- **Compact bone** – **Haversian system (osteon)**: Haversian canal (vessels), lamellae, lacunae (osteocytes), canaliculi; **Volkmann's canals** connect. **Spongy (cancellous)** – trabeculae, **red marrow**.
- **Cells** – **osteoblasts (bone forming)**, **osteocytes (mature)**, **osteoclasts (bone resorbing – multinucleated)**.
- **Ossification** – **intramembranous (flat skull bones, clavicle)**, **endochondral (from cartilage – long bones)**.
- **Red marrow in adults** – **flat bones (sternum, ilium, vertebrae, ribs, skull)**, proximal femur/ humerus; **bone marrow aspiration – posterior superior iliac spine (adult), sternum; tibia in infants**.

## 2.2 Axial Skeleton (80)

### Skull (22 bones) + hyoid + ear ossicles (6)

- **Cranial bones (8)** – **frontal (1), parietal (2), temporal (2), occipital (1), sphenoid (1), ethmoid (1)**.
- **Facial bones (14)** – **maxilla (2), zygomatic (2), nasal (2), lacrimal (2), palatine (2), inferior nasal concha (2), vomer (1), mandible (1 – only movable skull bone, largest & strongest facial bone)**.

- **Sutures** – **coronal** (frontal–parietal), **sagittal** (between parietals), **lambdoid** (parietal–occipital), squamous.
- **Foramen magnum** – in occipital – spinal cord/ medulla passes.
- **Ear ossicles** – **malleus, incus, stapes** (3 each side = 6).

### Fontanelles (newborn)

| Fontanelle | Shape | Closes |
|---|---|---|
| **Anterior (bregma)** | **Diamond**, largest (~2.5 × 3.5 cm) | **12–18 months** |
| **Posterior (lambda)** | **Triangular** | **6–8 weeks (by 2–3 months)** |

- **Bulging** – raised ICP (meningitis, hydrocephalus); **depressed/ sunken** – **dehydration**; delayed closure – **rickets, hypothyroidism, hydrocephalus**; early closure – craniosynostosis/ microcephaly.

### Vertebral column (26 in adult; 33 in child)

| Region | Number | Features |
|---|---|---|
| **Cervical** | **7** | **Foramen transversarium**; **C1 – atlas (no body, "yes" movement)**, **C2 – axis (dens/ odontoid process – "no" rotation)**, **C7 – vertebra prominens** |
| **Thoracic** | **12** | **Articulate with ribs** |
| **Lumbar** | **5** | **Largest bodies**; LP done at **L3–L4/ L4–L5** |
| **Sacrum** | 5 fused → 1 | |
| **Coccyx** | 4 fused → 1 | |

- **Curvatures** – **primary (concave anteriorly) – thoracic & sacral (fetal)**; **secondary – cervical (head holding ~3 months) & lumbar (walking ~12 months)**.
- **Abnormal** – **kyphosis (hunchback – ↑ thoracic)**, **lordosis (↑ lumbar – pregnancy)**, **scoliosis (lateral curvature – most common in adolescent girls)**.
- **Intervertebral disc** – **nucleus pulposus (inner, gel) + annulus fibrosus (outer)**; **disc prolapse commonest at L4–L5/ L5–S1**.

### Thoracic cage

- **Sternum** – **manubrium, body, xiphoid process**; **sternal angle (angle of Louis) – 2nd rib level** (counting ribs, T4/5); **CPR compressions – lower half of sternum**.
- **Ribs – 12 pairs (24)** – **true ribs 1–7** (directly to sternum), **false ribs 8–10** (via costal cartilage of 7th), **floating ribs 11–12**; **cervical rib** (extra – C7) → thoracic outlet syndrome.

## 2.3 Appendicular Skeleton (126)

| Part | Bones |
|---|---|
| **Pectoral girdle** | **Clavicle (collar bone – S-shaped, horizontal long bone with no medullary cavity)**, **scapula (shoulder blade)** |

| Part | Bones |
|---|---|
| **Upper limb (30 each)** | **Humerus**, **radius (lateral – thumb side)**, **ulna (medial; olecranon = elbow tip)**, **8 carpals** (**scaphoid – most commonly fractured carpal**; "**S**ome **L**overs **T**ry **P**ositions **T**hat **T**hey **C**an't **H**andle" – scaphoid, lunate, triquetrum, pisiform, trapezium, trapezoid, capitate, hamate), **5 metacarpals, 14 phalanges** |
| **Pelvic girdle** | **2 hip bones (ilium + ischium + pubis fused at acetabulum)**, sacrum, coccyx; **female pelvis – wider, shallower, round inlet (gynaecoid), subpubic angle > 90°**; male – android, narrow (< 70°) |
| **Lower limb (30 each)** | **Femur** (head, neck – **neck of femur fracture in elderly women – osteoporosis**), **patella**, **tibia (medial, weight-bearing; shin; medial malleolus)**, **fibula (lateral; lateral malleolus; non-weight bearing)**, **7 tarsals (calcaneus – largest, heel; talus – articulates with tibia)**, **5 metatarsals, 14 phalanges** |

- **Ischial tuberosity** – bone we sit on; **iliac crest** – highest point of hip (L4 level – landmark for LP); **anterior superior iliac spine (ASIS)**.
- **Arches of foot** – medial longitudinal (highest), lateral, transverse; **flat foot** – loss of arch.

## 2.4 Joints (Articulations)

| Type (structural) | Movement | Examples |
|---|---|---|
| **Fibrous** | **Immovable (synarthrosis)** | **Skull sutures**, gomphosis (**teeth in socket**), syndesmosis (distal tibiofibular) |
| **Cartilaginous** | **Slightly movable (amphiarthrosis)** | **Pubic symphysis, intervertebral discs** (secondary/ symphysis); **epiphyseal plate, 1st sternocostal** (primary/ synchondrosis) |
| **Synovial** | **Freely movable (diarthrosis)** | Most limb joints – **synovial membrane, synovial fluid, articular cartilage, capsule, ligaments**, bursae |

| Synovial type | Example |
|---|---|
| **Ball & socket** (most movement – multiaxial) | **Shoulder (most mobile, most commonly dislocated)**, **hip** |
| **Hinge** (uniaxial – flexion/ extension) | **Elbow, knee (largest joint – modified hinge), ankle, interphalangeal** |
| **Pivot** (rotation) | **Atlanto-axial (C1–C2 – "no")**, proximal radio-ulnar |
| **Condyloid (ellipsoid)** | **Wrist (radiocarpal)**, metacarpophalangeal |
| **Saddle** | **Carpometacarpal joint of thumb** (opposition) |
| **Gliding (plane)** | **Intercarpal, intertarsal**, acromioclavicular |

- **Movements** – **flexion** (↓ angle), **extension**, **hyperextension**, **abduction** (away from midline), **adduction** (towards), **rotation**, **circumduction**, **pronation** (palm down), **supination** (palm up – "soup"), **inversion/ eversion** (foot), **dorsiflexion/ plantar flexion**, **protraction/ retraction**, **opposition (thumb)**.
- **Knee** – **menisci (medial more often injured)**, **cruciate ligaments (ACL – sports injury)**, collateral ligaments; **largest & most complex joint**.
- **Ligament** – bone to bone; **tendon** – muscle to bone (**Achilles – strongest/ thickest tendon**).

## 2.5 Bone Physiology & Calcium

- **Normal serum calcium 8.5–10.5 mg/dl**; ↑ by **PTH (parathormone – bone resorption, renal reabsorption, activates vitamin D)** and **calcitriol (active vitamin D – 1,25-$(OH)_2D_3$, made in kidney; intestinal absorption)**; ↓ by **calcitonin (thyroid C cells – inhibits osteoclasts)**.
- **Vitamin D deficiency** – **rickets (children), osteomalacia (adults)**; **osteoporosis** – ↓ bone mass (post-menopausal – ↓ oestrogen; **DEXA scan; T-score ≤ −2.5**).
- **Bone healing stages** – haematoma → granulation/ inflammation → **soft callus** → **hard (bony) callus** → **remodelling**.
- **Growth** – GH, thyroxine, sex hormones (**puberty – epiphyseal fusion**), vitamin A, C (collagen), D.

**♫ Yaad karne ki trick**

- Vertebrae – **"Breakfast at 7, lunch at 12, dinner at 5"** – cervical 7, thoracic 12, lumbar 5.
- Cranial bones – **"PEST OF 6"** – **P**arietal, **E**thmoid, **S**phenoid, **T**emporal, **O**ccipital, **F**rontal (6 types, 8 bones).
- Fontanelle – **Anterior = A = Adult-like late (18 months)**; **Posterior = Pehle (2 months)**.
- Radius – **thumb side** ("**R**adius **R**uns to thumb").

**✎ PYQ style**

- Total adult bones – **206**; newborn – **~270–300**.
- Longest & strongest bone – **femur**; smallest – **stapes**.
- Most commonly fractured bone – **clavicle**; first to ossify – **clavicle**.
- Bone without articulation – **hyoid**.
- Only movable skull bone – **mandible**.
- Anterior fontanelle closes – **12–18 months**; posterior – **6–8 weeks**.
- Sunken fontanelle – **dehydration**; bulging – **raised ICP**.
- Cervical vertebrae – **7**; thoracic – **12**; lumbar – **5**.
- Atlas – **C1**; axis – **C2**; vertebra prominens – **C7**.
- LP site – **L3–L4 / L4–L5**; landmark – **iliac crest (L4)**.
- True ribs – **1–7**; floating – **11–12**.
- Largest sesamoid bone – **patella**; heel bone – **calcaneus**.
- Most commonly fractured carpal – **scaphoid**.
- Bone-forming cells – **osteoblasts**; resorbing – **osteoclasts**.
- Functional unit of compact bone – **osteon (Haversian system)**.
- Ball & socket – **shoulder, hip**; pivot – **atlanto-axial**; saddle – **thumb CMC**.
- Largest joint – **knee**; most commonly dislocated – **shoulder**.
- Immovable joint – **sutures**; pubic symphysis – **cartilaginous**.
- Tendon connects – **muscle to bone**; ligament – **bone to bone**.
- Calcitonin – **↓ serum calcium**; PTH – **↑ serum calcium**.

# 3 Muscular System

## 3.1 Types of Muscle

| Feature | Skeletal | Cardiac | Smooth |
|---|---|---|---|
| Control | **Voluntary** | **Involuntary** | **Involuntary** |
| Striations | **Present** | **Present** | **Absent** |
| Shape / nuclei | Long cylindrical, **multinucleated (peripheral)** | Branched, **1–2 central nuclei**, **intercalated discs** (gap junctions) | **Spindle-shaped, single central nucleus** |
| Nerve supply | Somatic | Autonomic (intrinsic – SA node) | Autonomic |
| Fatigue | Fatigues quickly | **Doesn't fatigue (long refractory period – no tetany)** | Slow, sustained |
| Location | Attached to bones | **Heart only** | **Walls of viscera, blood vessels, iris, uterus** |

- Body has **~600–650 skeletal muscles**; ≈ **40 % of body weight**.
- **Largest muscle – gluteus maximus**; **longest – sartorius**; **smallest – stapedius**; **strongest (by force/ weight) – masseter**; **largest smooth muscle organ – uterus (pregnancy)**; **most active – extraocular muscles**; **broadest – latissimus dorsi**; **hardest working – heart (myocardium)**.

## 3.2 Structure of Skeletal Muscle

- **Muscle → fascicles (perimysium) → muscle fibres/ cells (endomysium) → myofibrils → sarcomeres**; whole muscle covered by **epimysium**.
- **Sarcolemma** (cell membrane), **sarcoplasm**, **sarcoplasmic reticulum (stores $Ca^{2+}$)**, **T-tubules**.
- **Sarcomere – functional (contractile) unit** – between two **Z lines**; **thick filament – myosin (A band – dark, anisotropic)**, **thin filament – actin + troponin + tropomyosin (I band – light, isotropic)**; **H zone** (only myosin), **M line**.

### Neuromuscular junction & contraction

1. Nerve impulse → **acetylcholine (ACh)** released at **NMJ (motor end plate)**.
2. ACh binds **nicotinic receptors** → muscle action potential; ACh broken by **acetylcholinesterase**.
3. Impulse via T-tubules → **$Ca^{2+}$ released from sarcoplasmic reticulum**.
4. **$Ca^{2+}$ binds troponin** → tropomyosin shifts → actin binding sites exposed.
5. **Myosin heads (cross-bridges) bind actin & pull (power stroke) – ATP needed → sliding filament theory (Huxley)**: **I band & H zone shorten, A band length constant**.
6. Relaxation – $Ca^{2+}$ pumped back (ATP needed).
   - **Rigor mortis** – after death, **no ATP → myosin can't detach** (starts 2–6 hrs, lasts ~24–48 hrs).
   - **Myasthenia gravis** – **autoantibodies against ACh receptors** (ptosis, fatigue ↑ evening).

### Energy & physiology

- Sources – **creatine phosphate (first seconds)**, **anaerobic glycolysis → lactic acid (fatigue, cramps)**, **aerobic respiration (most ATP)**; **oxygen debt**; **myoglobin** – $O_2$ store in muscle (red fibres).
- **Motor unit** – one motor neuron + all fibres it supplies (eye – few fibres; thigh – many).
- **Muscle tone** – partial sustained contraction; **tetanus (physiological)** – sustained contraction from rapid stimuli; **all-or-none law** (single fibre).
- **Isotonic contraction** – length changes (concentric/ eccentric – lifting); **isometric** – tension ↑, length same (**pushing wall – isometric exercises for immobilised patients, quadriceps setting**).
- **Atrophy** (disuse, ↓ size), **hypertrophy** (exercise – ↑ size of fibres), **contracture**, **spasm, cramp**, **fibrillation**, **muscular dystrophy (Duchenne – X-linked, Gower's sign, boys)**.

## 3.3 Important Muscles

| Region | Muscle | Action / point |
|---|---|---|
| Face | **Orbicularis oculi** (closes eye), **orbicularis oris** (lips – "kissing muscle"), **buccinator** (cheek – blowing, sucking), frontalis | **Facial nerve (VII)** |
| Mastication | **Masseter, temporalis, pterygoids** | **Trigeminal (V3)**; masseter – strongest |
| Neck | **Sternocleidomastoid** (turns head to opposite side – **torticollis/ wry neck**), trapezius (shrug – **accessory XI**) | – |
| Respiration | **Diaphragm (main – 75 %, phrenic nerve C3–C5)**, **external intercostals (inspiration)**, internal intercostals (forced expiration); **accessory – SCM, scalene, pectoralis minor** (dyspnoea) | Abdominal muscles – forced expiration, cough |
| Abdomen | **Rectus abdominis** (linea alba, "six-pack"), external & internal oblique, transversus abdominis | – |
| Shoulder | **Deltoid** (abduction 15–90°; **IM site – axillary nerve**), **rotator cuff – SITS: supraspinatus (initiates abduction 0–15°), infraspinatus, teres minor, subscapularis** | – |
| Arm | **Biceps brachii** (flexion of elbow, supination), **triceps** (extension) | Antagonists |
| Back | **Latissimus dorsi** (broadest; "climbing"), erector spinae | – |
| Hip/ buttock | **Gluteus maximus** (extension – largest), **gluteus medius & minimus** (abduction; **ventrogluteal IM site**) | **Sciatic nerve** – avoid in dorsogluteal |
| Thigh | **Quadriceps femoris** (rectus femoris, vastus lateralis, medialis, intermedius – **knee extension**; **vastus lateralis – infant IM site**), **hamstrings** (knee flexion), **sartorius (longest – "tailor's muscle")**, adductors | – |
| Leg | **Gastrocnemius + soleus (calf – plantar flexion; Achilles tendon)**, tibialis anterior (dorsiflexion – foot drop if paralysed – common peroneal nerve) | – |
| Pelvic floor | **Levator ani** (pubococcygeus – **Kegel exercises**) | Supports pelvic organs |

### IM injection sites (anatomy link)

- **Deltoid** – **2–3 fingers (≈ 2.5–5 cm) below acromion**; **max 1 ml (≤ 2)**; avoid **radial & axillary nerve**.

- **Ventrogluteal** – **palm on greater trochanter, index on ASIS, middle finger along iliac crest – inject in V**; **safest (no major nerves/ vessels)**.
- **Dorsogluteal** – **upper outer quadrant** – risk of **sciatic nerve injury** (avoid in < 3 yrs).
- **Vastus lateralis** – **middle third of anterolateral thigh** – **site of choice in infants**.
- **Rectus femoris** – middle third anterior thigh (self-injection).

**Yaad karne ki trick**

- Rotator cuff – **"SITS"** – Supraspinatus, Infraspinatus, Teres minor, Subscapularis.
- Contraction – **"Calcium unlocks, ATP powers"**.
- Diaphragm nerve – **"C3, 4, 5 keeps the diaphragm alive"**.

**PYQ style**

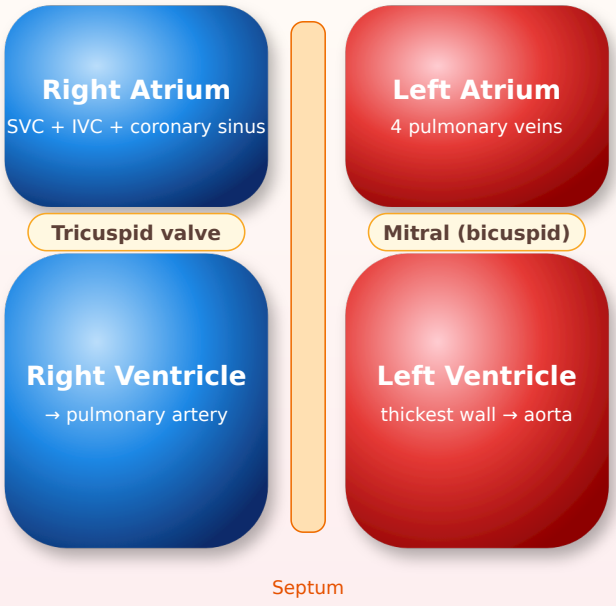
- Voluntary striated muscle – **skeletal**; intercalated discs – **cardiac**.
- Non-striated involuntary – **smooth**.
- Largest muscle – **gluteus maximus**; longest – **sartorius**; smallest – **stapedius**.
- Strongest muscle – **masseter**.
- Functional unit of muscle contraction – **sarcomere**.
- Thick filament – **myosin**; thin – **actin**.
- Ion triggering contraction – **calcium (binds troponin)**.
- Neurotransmitter at NMJ – **acetylcholine**.
- Sliding filament theory – **Huxley**; A band – **unchanged**.
- Rigor mortis cause – **ATP depletion**.
- Muscle fatigue – **lactic acid accumulation**.
- Main muscle of respiration – **diaphragm (phrenic C3–C5)**.
- Infant IM site – **vastus lateralis**; safest adult – **ventrogluteal**.
- Dorsogluteal risk – **sciatic nerve injury**.
- Kegel exercises – **pubococcygeus/ pelvic floor**.
- Foot drop – **common peroneal nerve** (tibialis anterior).
- Isometric exercise – **tension without length change**.
- Duchenne muscular dystrophy – **Gower's sign, X-linked**.

# 4 Cardiovascular System

## 4.1 The Heart – Anatomy

- **Location** – **middle mediastinum**, behind sternum, **⅔ to left of midline**; **apex – left 5th intercostal space, mid-clavicular line (~9 cm from midline)** – **apex beat/ PMI**; base – posterior-superior (great vessels).
- **Size** – ≈ **own clenched fist**; weight **~250–300 g (female ~250, male ~300)**.
- **Layers** – **pericardium** (outer fibrous + serous – parietal & visceral; **pericardial fluid 15–50 ml**; excess → **cardiac tamponade – Beck's triad: hypotension, muffled heart sounds, raised JVP**), **myocardium** (muscle – **thickest in left ventricle**), **endocardium** (inner – continuous with valves; infection → endocarditis).
- **Chambers** – **2 atria (receiving)**, **2 ventricles (pumping)**; **interatrial septum** (fossa ovalis – remnant of foramen ovale), **interventricular septum**.

### Heart – Chambers, Valves & Blood Flow




**Blood flow pathway**

1. Body → SVC / IVC
2. Right atrium → tricuspid
3. Right ventricle → pulmonary valve
4. Pulmonary artery (deoxygenated)
5. Lungs (gas exchange)
6. Pulmonary veins (oxygenated)
7. Left atrium → mitral valve
8. Left ventricle → aortic valve
9. Aorta → whole body

**Conduction: SA node (pacemaker 60–100)** → AV node (40–60) → Bundle of His → bundle branches → Purkinje (20–40)

Sounds: S1 "lub" (AV close) · S2 "dub" (semilunar)

Blue = deoxygenated (right side) · Red = oxygenated (left side)

*Heart – chambers, valves, blood flow pathway & conduction system*

| Valve | Location | Cusps |
|---|---|---|
| **Tricuspid** | Right atrium → right ventricle | 3 |
| **Mitral (bicuspid)** | Left atrium → left ventricle | **2 (only valve with 2 cusps)** |
| **Pulmonary (semilunar)** | RV → pulmonary trunk | 3 |
| **Aortic (semilunar)** | LV → aorta | 3 |

- **AV valves** held by **chordae tendineae + papillary muscles** (prevent eversion); semilunar valves have no chordae.
- **Valves prevent backflow**; **most commonly affected in rheumatic heart disease – mitral**.
- **Pulmonary artery – only artery carrying deoxygenated blood**; **pulmonary veins (4) – only veins carrying oxygenated blood** (in adults); **umbilical vein** – oxygenated (fetus).

### Coronary circulation

- **Right & left coronary arteries** arise from **aortic sinuses (just above aortic valve)**; **LCA → left anterior descending (LAD – "widow maker", most commonly blocked) + circumflex**; **RCA – supplies SA node (~60 %) & AV node (~80–90 %)**, inferior wall.
- Coronary filling occurs mainly during **diastole**.
- Venous drainage – **coronary sinus → right atrium**.

## 4.2 Conduction System & Cardiac Cycle

| Structure | Intrinsic rate |
|---|---|
| **SA node (pacemaker)** – right atrium wall near SVC | **60–100/min** |
| **AV node** – interatrial septum (delay ~0.1 s – allows ventricular filling) | **40–60/min** |
| **Bundle of His → right & left bundle branches** | – |
| **Purkinje fibres** | **20–40/min** |

- **Cardiac muscle – autorhythmicity**; **long refractory period → no tetany**; **Frank-Starling law** – ↑ stretch (preload) → ↑ force of contraction.
- **Cardiac cycle ≈ 0.8 s at 75 bpm** – **atrial systole 0.1 s, ventricular systole 0.3 s, diastole 0.5 s** (diastole shortens with tachycardia).
- **Heart sounds** – **S1 "lub" – closure of AV (mitral & tricuspid) valves – start of systole (loudest at apex)**; **S2 "dub" – closure of semilunar (aortic & pulmonary) – end of systole**; **S3** – rapid ventricular filling (normal in children/ pregnancy; **heart failure** in adults); **S4** – atrial contraction against stiff ventricle (always abnormal).
- **Auscultation areas** – **aortic: 2nd right ICS at sternal border**; **pulmonary: 2nd left ICS**; **Erb's point: 3rd left ICS**; **tricuspid: 4th–5th left ICS at sternal border**; **mitral: 5th left ICS mid-clavicular line (apex)** – "**All Physicians Eat Too Much**".
- **ECG** – **P wave – atrial depolarisation**; **PR interval 0.12–0.20 s (AV conduction)**; **QRS – ventricular depolarisation (< 0.12 s)** (atrial repolarisation hidden); **T wave – ventricular repolarisation**; ST segment – elevation in MI; **U wave – hypokalaemia**; **tall peaked T – hyperkalaemia**; **standard 12 leads**; paper speed **25 mm/s – small square 0.04 s, large square 0.2 s**; heart rate = **300 ÷ number of large squares between R–R** (or 1500 ÷ small squares).

## 4.3 Cardiac Output & Blood Pressure

- **Stroke volume (SV) ≈ 70 ml**; **Cardiac output (CO) = SV × HR ≈ 70 × 72 ≈ 5 L/min**; **cardiac index ≈ 3 L/min/m²**; **ejection fraction (EF) = SV/EDV ≈ 55–70 %** (< 40 % – HFrEF); **EDV ≈ 120 ml, ESV ≈ 50 ml**.
- **Preload** – venous return/ EDV; **afterload** – resistance against which ventricle pumps (aortic pressure, SVR); **contractility** – inotropy.

- **Blood pressure = CO × total peripheral resistance (TPR)**; **normal adult ~120/80 mmHg**; **pulse pressure = SBP − DBP (≈ 40)**; **MAP = DBP + ⅓ pulse pressure (≈ 93; normal 70–105; ≥ 65 for organ perfusion)**.
- **Regulation** – **short-term: baroreceptors (carotid sinus & aortic arch – glossopharyngeal & vagus)**, chemoreceptors, **vasomotor centre (medulla)**; **long-term: kidney – RAAS (renin → angiotensin I → ACE (lungs) → angiotensin II – vasoconstriction & aldosterone)**, ADH, ANP (atria – natriuresis, ↓ BP).
- **Factors ↑ BP** – exercise, emotion, ↑ blood volume, ↑ viscosity, age, obesity; **↓ BP** – sleep, haemorrhage, vasodilation.
- **Korotkoff sounds** – **phase I – SBP (first tapping)**, **phase V – DBP (disappearance)**; cuff bladder **width ≈ 40 %, length ≈ 80 % of arm circumference**; small cuff → **falsely high**; large cuff → falsely low.
- **Normal pulse** – **adult 60–100**, newborn 120–160, child 80–120; **bradycardia < 60**, **tachycardia > 100**; **apical-radial pulse deficit (AF)**.

## 4.4 Blood Vessels

| Vessel | Features |
|---|---|
| **Arteries** | Carry blood **away from heart**; **thick muscular tunica media**; elastic (aorta) & muscular; **arterioles – main resistance vessels (regulate BP)** |
| **Capillaries** | **Single layer of endothelium** – **exchange vessels**; slowest flow; largest total cross-section |
| **Veins** | Towards heart; **thin walls, large lumen, valves (prevent backflow – esp. lower limbs)**; **capacitance vessels – hold ~60–70 % of blood volume** |

- **Wall layers** – **tunica intima (endothelium), tunica media (smooth muscle – thickest in arteries), tunica adventitia/ externa (thickest in veins)**.
- **Largest artery – aorta**; **largest vein – inferior vena cava**; **portal vein** – GI tract → liver (**hepatic portal system** – vein between two capillary beds; also **hypothalamo-hypophyseal portal**).
- **Aorta** – ascending (coronaries) → **arch** (**brachiocephalic trunk → right subclavian + right common carotid; left common carotid; left subclavian**) → descending thoracic → abdominal (coeliac trunk, SMA, renal, IMA) → **bifurcates at L4 into common iliacs**.
- **Circle of Willis** – arterial anastomosis at base of brain (internal carotids + vertebrals/ basilar).
- **Veins for cannulation** – **cephalic, basilic, median cubital (venepuncture – antecubital fossa)**, dorsal venous arch of hand; **great saphenous vein – longest vein (CABG graft; varicose veins)**; **jugular – JVP (normal ≤ 3–4 cm above sternal angle at 45°)**.

### Pulse points

- **Radial** (commonest; wrist, thumb side), **carotid** (**CPR in adults/ children > 1 yr**; never both sides), **brachial** (**infants – CPR**; BP), **femoral**, popliteal (behind knee), **posterior tibial** (behind medial malleolus), **dorsalis pedis** (dorsum of foot), temporal, **apical** (infants < 2 yrs – count 1 full minute; digoxin).

## 4.5 Fetal Circulation

- **Umbilical cord – 2 arteries (deoxygenated → placenta) + 1 vein (oxygenated → fetus)** ("**AVA**"); Wharton's jelly.

| Fetal shunt | Connects | After birth becomes |
|---|---|---|
| **Ductus venosus** | Umbilical vein → IVC (bypasses liver) | **Ligamentum venosum** |
| **Foramen ovale** | Right atrium → left atrium (bypasses lungs) | **Fossa ovalis** (functional closure – first breath, ↑ LA pressure) |
| **Ductus arteriosus** | Pulmonary artery → aorta (bypasses lungs) | **Ligamentum arteriosum** (closes 24–72 hrs functionally; **PDA if open – indomethacin/ ibuprofen closes, PGE1 keeps open**) |
| **Umbilical vein** | – | **Ligamentum teres (round ligament of liver)** |
| **Umbilical arteries** | – | **Medial umbilical ligaments** |

- **Highest $O_2$ saturation – umbilical vein (~80 %)**; fetal Hb (HbF) has **higher $O_2$ affinity**.

### ♫ Yaad karne ki trick

- Valves order – **"Try Pulling My Aorta"** – Tricuspid, Pulmonary, Mitral, Aortic.
- Auscultation – **"All Physicians Eat Too Much"** – Aortic, Pulmonary, Erb's, Tricuspid, Mitral.
- Umbilical cord – **"AVA"** – 2 arteries, 1 vein.
- Mitral = Bicuspid = **Left** – "**LMB – Left Mitral Bicuspid**".

### ✎ PYQ style

- Pacemaker of heart – **SA node (60–100/min)**.
- Thickest chamber – **left ventricle**.
- Valve with 2 cusps – **mitral**; RHD commonly affects – **mitral**.
- Artery carrying deoxygenated blood – **pulmonary artery**; vein with oxygenated blood – **pulmonary vein**.
- Apex beat – **left 5th ICS, mid-clavicular line**.
- S1 – **closure of AV valves**; S2 – **closure of semilunar valves**.
- Cardiac cycle duration – **0.8 s**.
- Cardiac output – **≈ 5 L/min**; stroke volume – **≈ 70 ml**.
- Normal ejection fraction – **55–70 %**.
- P wave – **atrial depolarisation**; T wave – **ventricular repolarisation**.
- Normal PR interval – **0.12–0.20 s**.
- Coronary filling – **during diastole**; most commonly blocked – **LAD**.
- Largest artery – **aorta**; largest vein – **IVC**; longest vein – **great saphenous**.
- Resistance vessels – **arterioles**; exchange – **capillaries**; capacitance – **veins**.
- Baroreceptors – **carotid sinus & aortic arch**.
- First Korotkoff sound – **systolic BP**.
- Infant CPR pulse – **brachial**; adult – **carotid**.
- Foramen ovale becomes – **fossa ovalis**; ductus arteriosus – **ligamentum arteriosum**.
- Umbilical cord – **2 arteries, 1 vein**.
- Cardiac tamponade – **Beck's triad**.

# 5 Blood, Lymphatic System & Immunity

## 5.1 Blood – General

- **Blood volume** – **~5–6 L in adult (≈ 7–8 % body weight; 70–80 ml/kg)**; **newborn ~80–85 ml/kg**; **pH 7.35–7.45**; specific gravity ~1.055; viscosity 4–5 times water.
- **Composition** – **plasma ≈ 55 %** + **formed elements ≈ 45 % (haematocrit/ PCV)**.
- **Plasma** – **91–92 % water**, proteins **7 % (6–8 g/dl)** – **albumin (3.5–5 g/dl; most abundant – oncotic pressure, transport; made in liver)**, **globulins (antibodies – γ-globulins, transport)**, **fibrinogen (clotting)**; **serum = plasma − clotting factors (fibrinogen)**.
- **Functions** – transport ($O_2$, $CO_2$, nutrients, hormones, waste), **protection (immunity, clotting)**, **regulation (pH buffer, temperature, fluid balance)**.
- **Haemopoiesis** – **fetus: yolk sac (3 wks) → liver (main, 2nd trimester) & spleen → bone marrow (from ~5th month)**; **adult – red marrow of flat bones**; all from **pluripotent haemopoietic stem cell**.

## 5.2 Red Blood Cells (Erythrocytes)

- **Biconcave disc, non-nucleated**, diameter **~7.5 µm**; **lifespan 120 days**; destroyed in **spleen ("graveyard of RBCs")**, liver, marrow.
- **Count – male 4.5–5.5 million/µl, female 4–5 million/µl**.
- **Erythropoiesis** – stimulated by **erythropoietin (kidney – in hypoxia)**; needs **iron, vitamin B12, folic acid**, protein, vitamin C, copper; stages – proerythroblast → … → **reticulocyte (0.5–2.5 %)** → RBC (~7 days).
- **Haemoglobin** – **male 13–17 g/dl, female 12–15 g/dl, newborn 15–20 g/dl**; **pregnancy anaemia < 11 g/dl**; **4 haem (iron $Fe^{2+}$) + globin (2α + 2β in HbA)**; **1 g Hb carries 1.34 ml $O_2$**; **HbF (2α2γ) – higher $O_2$ affinity**; **HbS – sickle cell (valine replaces glutamic acid)**.
- **Breakdown** – Hb → haem → **biliverdin → bilirubin (unconjugated, fat-soluble) → liver conjugates (glucuronyl transferase) → bile → stercobilinogen (stool colour) & urobilinogen (urine)**; iron recycled (ferritin/ transferrin).
- **Indices** – **MCV 80–100 fl** (↓ **microcytic – iron deficiency, thalassaemia**; ↑ **macrocytic – B12/ folate deficiency**), MCH 27–32 pg, MCHC 32–36 g/dl; **ESR – Westergren (male 0–15, female 0–20 mm/hr)**.
- **Jaundice** when **serum bilirubin > 2–3 mg/dl** (normal 0.3–1.2); **newborn visible > 5 mg/dl**.

## 5.3 White Blood Cells (Leucocytes)

- **Total count – 4,000–11,000/µl**; **leucocytosis > 11,000**; **leucopenia < 4,000**.

| Type | % (DLC) | Function | ↑ in |
|---|---|---|---|
| **Neutrophils** (multilobed nucleus) | **50–70 % (most common)** | **First line – phagocytosis (pus)**, lifespan 6–8 hrs in blood | **Acute bacterial infection** |
| **Lymphocytes** | **20–40 %** | **Immunity – B (antibodies), T (cell-mediated), NK cells** | **Viral infections, TB**, CLL |

| Type | % (DLC) | Function | ↑ in |
|---|---|---|---|
| **Monocytes** (largest WBC) | 2–8 % | **Become macrophages** in tissues (Kupffer cells – liver, microglia – brain, alveolar macrophages) | TB, chronic infection |
| **Eosinophils** (bilobed, red granules) | 1–4 % | **Allergy, parasites** | **Allergy, worm infestation, asthma** |
| **Basophils** (least common) | 0–1 % | **Histamine, heparin** (like mast cells) | Allergy, CML |

- **Granulocytes** – neutrophil, eosinophil, basophil; **agranulocytes** – lymphocyte, monocyte.
- "**Never Let Monkeys Eat Bananas**" – order of frequency (N > L > M > E > B).

## 5.4 Platelets & Haemostasis

- **Platelets (thrombocytes)** – fragments of **megakaryocytes**; **1.5–4.5 lakh/µl**; lifespan **7–10 days**; **thrombocytopenia < 1.5 lakh** (bleeding risk < 50,000; spontaneous < 20,000); **thrombopoietin (liver)**.
- **Haemostasis steps** – **1. vascular spasm → 2. platelet plug (adhesion via vWF, aggregation – ADP, TXA2) → 3. coagulation (fibrin clot) → 4. clot retraction → 5. fibrinolysis (plasmin)**.
- **Coagulation** – **intrinsic pathway (aPTT – factors XII, XI, IX, VIII; heparin)**, **extrinsic pathway (PT/INR – tissue factor + VII; warfarin) → common pathway (X, V, II – prothrombin → thrombin; I – fibrinogen → fibrin; XIII stabilises)**.
- **Calcium = factor IV** (needed throughout; **citrate/ EDTA/ oxalate anticoagulants bind calcium**); **vitamin K-dependent factors – II, VII, IX, X** ("**1972**") + protein C & S – made in liver.
- **Normal values** – **bleeding time 2–7 min (Ivy)** (platelet function), **clotting time 4–10 min**, **PT 11–16 s, INR ~1**, **aPTT 25–35 s**.
- **Haemophilia A – factor VIII deficiency (X-linked recessive, males)**; **B (Christmas disease) – factor IX**; **von Willebrand disease – commonest inherited bleeding disorder**.
- **Natural anticoagulants** – **heparin (basophils, mast cells), antithrombin III, protein C & S**.

## 5.5 Blood Groups & Transfusion

- **ABO system – Karl Landsteiner (1900–01)**; **Rh – Landsteiner & Wiener (1940)**; **World Blood Donor Day – 14 June** (Landsteiner's birthday).

| Group | Antigen on RBC | Antibody in plasma | Can donate to | Can receive from |
|---|---|---|---|---|
| **A** | A | Anti-B | A, AB | A, O |
| **B** | B | Anti-A | B, AB | B, O |
| **AB** | A & B | **None** | AB | **All – universal recipient** |
| **O** | **None** | Anti-A & anti-B | **All – universal donor (RBC)** | O |

- **Plasma** – **AB – universal plasma donor**; **O – universal plasma recipient**.
- **Commonest blood groups in India – O+ and B+ (varies by region; B+ more in north)**; **Rh-positive ≈ 93–95 % in India**; **Bombay blood group (hh) – no H antigen** (first found in Mumbai, 1952).
- **Rh incompatibility** – **Rh−ve mother + Rh+ve fetus** → sensitisation (first pregnancy usually unaffected) → **haemolytic disease of newborn/ erythroblastosis fetalis** in later pregnancies →

**anti-D immunoglobulin (300 µg) within 72 hrs of delivery/ abortion & at 28 weeks**.

- **Donation** – **age 18–65 yrs, weight ≥ 45 kg (350 ml) / ≥ 55 kg (450 ml), Hb ≥ 12.5 g/dl**; interval **3 months (men), 4 months (women)**; whole blood stored **2–6 °C**, with **CPDA-1 – 35 days** (SAGM – 42 days); **platelets 20–24 °C with agitation – 5 days**; **FFP −30 °C or below – 1 year**.
- **Cross-matching** – donor RBC + recipient serum (major); **Coombs test** – direct (antibody on RBC – HDN, AIHA), indirect (antibodies in serum – antenatal).

## 5.6 Lymphatic System

- **Functions** – return of **tissue fluid (~3 L/day) & proteins** to blood, **absorption of fats (lacteals – chyle)**, **immunity (lymphocytes)**.
- **Lymph flow** – capillaries → vessels (valves) → **nodes (filter)** → trunks → **thoracic duct (largest; drains left side + both lower limbs; starts at cisterna chyli, enters left subclavian–internal jugular junction)** & **right lymphatic duct (right upper quadrant)**.
- **Primary lymphoid organs** – **bone marrow (B cells mature)**, **thymus (T cells mature – large in children, involutes after puberty; "T" for thymus)**; **secondary** – lymph nodes, **spleen**, tonsils, **Peyer's patches (ileum)**, MALT, appendix.
- **Spleen** – **largest lymphoid organ**; **left hypochondrium, ribs 9–11**; **not normally palpable**; functions – **destroys old RBCs, filters blood, immune response, reservoir of blood/ platelets, haemopoiesis in fetus**; **post-splenectomy – risk of overwhelming infection (encapsulated – pneumococcus, H. influenzae, meningococcus) → vaccinate**; **ruptured spleen – Kehr's sign (left shoulder pain)**.
- **Tonsils** – palatine (most common "tonsils"), pharyngeal (**adenoids**), lingual.
- **Lymph nodes** – cervical, axillary (**breast cancer spread**), inguinal; **Virchow's node – left supraclavicular (gastric cancer – Troisier's sign)**.

## 5.7 Immunity

| Type | Details |
|---|---|
| **Innate (non-specific, natural)** | **Skin & mucous membranes (first line)**, **phagocytes (neutrophils, macrophages), NK cells, complement, inflammation, fever, interferons (second line)**; no memory |
| **Acquired (adaptive, specific)** | **Third line** – lymphocytes; **memory & specificity** |
| – **Humoral** | **B lymphocytes → plasma cells → antibodies** |
| – **Cell-mediated** | **T lymphocytes – helper (CD4 – HIV target), cytotoxic (CD8), regulatory, memory** – viruses, TB, transplant rejection, cancer, **type IV hypersensitivity (Mantoux)** |

| Acquired immunity | Active (own antibodies, long-lasting, delayed) | Passive (ready-made antibodies, immediate, short) |
|---|---|---|
| **Natural** | **Recovery from infection** | **Maternal antibodies – placenta (IgG) & breast milk/ colostrum (IgA)** |
| **Artificial** | **Vaccination** | **Antisera/ immunoglobulins (ATS, ASV, RIG, HBIG)** |

## Antibodies (immunoglobulins – Y-shaped; plasma cells)

| Ig | Key point |
|---|---|
| **IgG** | **Most abundant (~75 %)**; **only one crossing placenta**; secondary response; long-term |
| **IgM** | **Largest (pentamer)**; **first produced in primary response** (recent infection); ABO antibodies |
| **IgA** | **Secretions – colostrum, breast milk, saliva, tears, mucosa (dimer)** – secretory immunity |
| **IgE** | **Allergy/ anaphylaxis (type I), parasites**; binds mast cells |
| **IgD** | B-cell receptor |

- **Antigen** – substance inducing immune response; **hapten** – incomplete antigen; **vaccine** – live attenuated (BCG, OPV, MR, rotavirus), killed (IPV), toxoid (TT, DT), subunit/ recombinant (hepatitis B).
- **Hypersensitivity** – **Type I – anaphylactic (IgE – asthma, hay fever, anaphylaxis)**; **Type II – cytotoxic (IgG/ IgM – transfusion reaction, HDN, Graves', myasthenia)**; **Type III – immune complex (serum sickness, SLE, PSGN, Arthus)**; **Type IV – delayed, T-cell (Mantoux, contact dermatitis, graft rejection)** – "**ACID**".
- **Autoimmunity** – loss of self-tolerance (SLE, RA, type 1 DM, Hashimoto's).

**Yaad karne ki trick**

- WBC frequency – **"Never Let Monkeys Eat Bananas"**.
- Vitamin K factors – **"1972"** → **10, 9, 7, 2** (X, IX, VII, II).
- Hypersensitivity – **"ACID"** – Anaphylactic, Cytotoxic, Immune complex, Delayed.
- Ig crossing placenta – **"G for Gestation"** – IgG.

### PYQ style

- Blood volume adult – **5–6 L**; pH – **7.35–7.45**.
- Plasma – **55 %**; most abundant plasma protein – **albumin**.
- RBC lifespan – **120 days**; graveyard of RBC – **spleen**.
- Erythropoietin secreted by – **kidney**.
- Normal Hb female – **12–15 g/dl**; male – **13–17 g/dl**.
- Most common WBC – **neutrophil**; largest – **monocyte**; least – **basophil**.
- ↑ eosinophils – **allergy, worm infestation**.
- Normal platelet count – **1.5–4.5 lakh/µl**; lifespan – **7–10 days**.
- Factor IV – **calcium**; vitamin K-dependent – **II, VII, IX, X**.
- Haemophilia A – **factor VIII deficiency**.
- Normal bleeding time – **2–7 min**; clotting time – **4–10 min**.
- Universal donor – **O −ve**; universal recipient – **AB +ve**.
- ABO discovered by – **Karl Landsteiner**.
- Anti-D given within – **72 hrs**.
- Largest lymphoid organ – **spleen**; T-cell maturation – **thymus**.
- Largest lymph vessel – **thoracic duct**.
- Antibody crossing placenta – **IgG**; in colostrum – **IgA**; first formed – **IgM**; allergy – **IgE**.
- HIV attacks – **CD4 T-helper cells**.
- Vaccination – **artificial active immunity**; antiserum – **artificial passive**.
- Mantoux – **type IV hypersensitivity**.

# 6 Respiratory System

## 6.1 Anatomy of Respiratory Tract

- **Upper tract** – nose, nasal cavity, paranasal sinuses, pharynx, larynx; **lower tract** – trachea, bronchi, bronchioles, alveoli (lungs).
- **Conducting zone** (nose → terminal bronchioles – **anatomical dead space ≈ 150 ml**); **respiratory zone** (respiratory bronchioles → alveolar ducts → **alveoli – gas exchange**).
- **Nose** – warms, humidifies, filters air (hairs, mucus, cilia); **nasal septum** (cartilage + vomer + ethmoid); **Little's area (Kiesselbach's plexus) – anterior septum – commonest site of epistaxis**; **olfactory epithelium** – roof.
- **Paranasal sinuses** – **maxillary (largest; commonest sinusitis; present at birth)**, frontal, ethmoid, sphenoid; lighten skull, voice resonance.
- **Pharynx** (~12–14 cm) – **nasopharynx (adenoids, opening of Eustachian tube)**, **oropharynx (palatine tonsils)**, **laryngopharynx** – **common passage for air & food**.
- **Larynx (voice box)** – **C3–C6**; **9 cartilages** – **thyroid (largest – Adam's apple)**, **cricoid (only complete ring – narrowest part of airway in children < 8 yrs)**, **epiglottis (elastic – closes larynx during swallowing)**, paired arytenoid, corniculate, cuneiform; **vocal cords**; **glottis – narrowest part in adults**; **nerve – recurrent laryngeal (injury in thyroidectomy → hoarseness)**.
- **Trachea (windpipe)** – **~10–12 cm, from C6 to T4/T5 (sternal angle)**; **16–20 C-shaped hyaline cartilage rings** (open posteriorly – oesophagus expands); **carina – bifurcation (most sensitive area – cough reflex)**; lined by **pseudostratified ciliated columnar epithelium with goblet cells – mucociliary escalator**.
- **Bronchi** – **right main bronchus – shorter, wider, more vertical → foreign bodies & aspiration commonly enter right side (right lower lobe)**; bronchi → bronchioles (no cartilage, smooth muscle – bronchospasm) → terminal → respiratory bronchioles.



*Respiratory tree, lobes of lungs & the alveolus*

## 6.2 Lungs & Pleura

| Feature | Right lung | Left lung |
|---|---|---|
| Lobes | **3 (upper, middle, lower)** | **2 (upper, lower)** |
| Fissures | **2 (oblique + horizontal)** | **1 (oblique)** |
| Size | **Larger, heavier, shorter, wider** | Smaller, longer, narrower; **cardiac notch, lingula** |
| Bronchopulmonary segments | 10 | 8–10 |

- **Apex** – rises **2–3 cm above clavicle** (into neck – risk in subclavian cannulation → pneumothorax); **base** – on diaphragm; **hilum** – root (bronchus, pulmonary vessels, nerves, lymphatics).
- **Pleura** – **visceral (on lung – no pain fibres)** & **parietal (lines chest wall – pain sensitive)**; **pleural cavity with ~10–20 ml serous fluid** – lubrication; **negative intrapleural pressure (≈ −4 to −5**

**cmH₂O)** keeps lungs expanded; **air → pneumothorax, fluid → pleural effusion, pus → empyema, blood → haemothorax**.

- **Blood supply** – **pulmonary circulation (gas exchange)** + **bronchial arteries (from aorta – nourish lung tissue)**.

### Alveoli

- **~300 million (300–500 million) alveoli**; **surface area ≈ 70 m²** (tennis court ~ 70–100 m²); **respiratory membrane ≈ 0.5 µm thick**.
- **Type I pneumocytes – ~95 % surface, gas exchange**; **type II – secrete surfactant**; **alveolar macrophages (dust cells)**.
- **Surfactant (dipalmitoyl phosphatidylcholine/ lecithin)** – **↓ surface tension → prevents alveolar collapse (atelectasis)**; production starts **~24–28 weeks, adequate by ~34–35 weeks**; deficiency → **RDS (hyaline membrane disease) in preterm** → **antenatal corticosteroids, exogenous surfactant**; **L/S ratio ≥ 2 – lung maturity**.

## 6.3 Mechanics of Breathing

- **Inspiration – active** – **diaphragm contracts & descends (main – ~75 %)**, **external intercostals** raise ribs → **↑ thoracic volume → ↓ intrapulmonary pressure (Boyle's law) → air enters**.
- **Expiration – passive (quiet)** – elastic recoil; **forced expiration – active** (internal intercostals, abdominal muscles).
- **Accessory muscles** – SCM, scalenes, pectoralis (dyspnoea, COPD).
- **Compliance** – ease of lung expansion (↓ in fibrosis, RDS; ↑ in emphysema); **airway resistance** (↑ in asthma).

### Lung volumes & capacities (adult male approx.)

| Volume | Value | Definition |
|---|---|---|
| **Tidal volume (TV)** | **≈ 500 ml** | Normal quiet breath |
| **Inspiratory reserve volume (IRV)** | ≈ 3000 ml | Extra inspiration after normal |
| **Expiratory reserve volume (ERV)** | ≈ 1100–1200 ml | Extra expiration |
| **Residual volume (RV)** | **≈ 1200 ml** | **Remains after maximal expiration – can't be measured by spirometry** |
| **Vital capacity (VC)** | **≈ 4500–4800 ml** | **TV + IRV + ERV** (max air expelled after max inspiration) |
| **Total lung capacity (TLC)** | **≈ 5800–6000 ml** | VC + RV |
| **Functional residual capacity (FRC)** | ≈ 2300–2400 ml | ERV + RV |
| **Inspiratory capacity (IC)** | ≈ 3500 ml | TV + IRV |

- **Minute ventilation = TV × RR ≈ 500 × 12 = 6 L/min**; **alveolar ventilation = (TV − dead space) × RR ≈ 4.2 L/min**.

- **FEV1/FVC** – normal **≥ 70–80 %**; **↓ in obstructive (asthma, COPD)**; normal/ ↑ in restrictive (FVC ↓).
- **Peak expiratory flow (PEF)** – peak flow meter – asthma monitoring.
- **Normal RR** – **adult 12–20/min**, **newborn 30–60/min**, infant 30–50, child 20–30; ratio **inspiration : expiration ≈ 1 : 2**.

## 6.4 Gas Exchange & Transport

- **Diffusion** – partial pressure gradient (**Dalton's law**); **$CO_2$ diffuses ~20 times faster than $O_2$** (more soluble).
- **Partial pressures** – **alveolar $PO_2$ ≈ 100 mmHg, $PCO_2$ ≈ 40**; **arterial $PaO_2$ 80–100, $PaCO_2$ 35–45 mmHg**; **venous $PO_2$ ≈ 40, $PCO_2$ ≈ 46**; **$SpO_2$ normal 95–100 %**.
- **$O_2$ transport** – **~97–98.5 % bound to haemoglobin (oxyhaemoglobin)**, **~1.5–3 % dissolved in plasma**.
- **Oxygen-haemoglobin dissociation curve – sigmoid (S-shaped)**; **shift to RIGHT (↓ affinity – $O_2$ released to tissues)** – **↑ $CO_2$, ↑ $H^+$ (↓ pH – Bohr effect), ↑ temperature, ↑ 2,3-DPG, exercise**; **shift to LEFT** – ↓ temperature, alkalosis, **HbF, carbon monoxide (CO affinity ~200–250 × $O_2$)**; **P50 ≈ 27 mmHg**.
- **$CO_2$ transport** – **~70 % as bicarbonate ($HCO_3^-$ – carbonic anhydrase in RBC; chloride shift/ Hamburger phenomenon)**, **~20–23 % as carbaminohaemoglobin**, **~7–10 % dissolved**; **Haldane effect** – deoxygenated Hb carries more $CO_2$.

## 6.5 Control of Respiration

- **Respiratory centres** – **medulla oblongata (dorsal – inspiratory, ventral respiratory group; rhythm generator – pre-Bötzinger complex)**, **pons (pneumotaxic – limits inspiration; apneustic)**.
- **Central chemoreceptors (medulla)** – **most sensitive to ↑ $CO_2$ / $H^+$ in CSF (main drive)**; **peripheral chemoreceptors (carotid & aortic bodies)** – respond to **↓ $O_2$ ($PaO_2$ < 60)**, ↑ $CO_2$, ↓ pH; **COPD with chronic $CO_2$ retention – hypoxic drive** (controlled $O_2$ – target $SpO_2$ 88–92 %).
- **Hering-Breuer reflex** – stretch receptors prevent over-inflation (vagus).
- **Cough reflex** (vagus), **sneeze**, **hiccup** (diaphragm spasm – phrenic).

### Terms & abnormal patterns

| Term | Meaning |
|---|---|
| **Eupnoea** | Normal breathing |
| **Tachypnoea / bradypnoea** | **> 20 / < 12 per min** (adult) |
| **Apnoea** | Absence of breathing (**> 20 s in newborn**) |
| **Dyspnoea / orthopnoea** | Difficulty / **difficulty lying flat (heart failure – sleeps on pillows)**; **PND** – paroxysmal nocturnal dyspnoea |
| **Hyperpnoea / hyperventilation** | ↑ depth / ↑ rate & depth (↓ $CO_2$ – respiratory alkalosis) |
| **Kussmaul's breathing** | **Deep, rapid, laboured – metabolic acidosis (DKA)** – acetone breath |
| **Cheyne-Stokes** | **Cyclic waxing-waning with apnoea – heart failure, raised ICP, dying, brainstem lesion** |
| **Biot's (ataxic)** | Irregular with apnoea – **meningitis, medulla damage** |

| Term | Meaning |
|---|---|
| **Apneustic** | Prolonged inspiratory gasps – pons lesion |
| **Hypoxia / hypoxaemia** | ↓ $O_2$ in tissues / in blood; **types – hypoxic, anaemic, stagnant (circulatory), histotoxic (cyanide)** |
| **Cyanosis** | Bluish discolouration when **deoxygenated Hb > 5 g/dl**; central (tongue, lips) vs peripheral; may be absent in severe anaemia |
| **Asphyxia** | Hypoxia + hypercapnia |

### ♫ Yaad karne ki trick

- Right bronchus – **"Right is Rough route"** – shorter, wider, vertical → foreign body.
- Right shift of curve – **"CADET face Right"** – **C**$O_2$, **A**cid, **D**PG, **E**xercise, **T**emperature.
- Lobes – **Right 3, Left 2** (left gives space to heart – cardiac notch).

### ✎ PYQ style

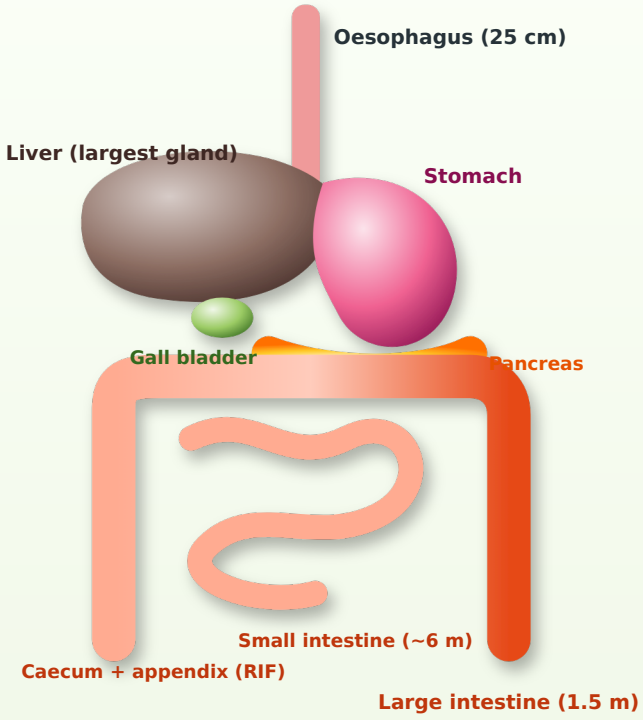
- Commonest site of epistaxis – **Little's area**.
- Largest paranasal sinus – **maxillary**.
- Largest laryngeal cartilage – **thyroid**; only complete ring – **cricoid**.
- Narrowest part of airway in child – **cricoid**; adult – **glottis**.
- Trachea bifurcates at – **T4/T5 (carina)**.
- Foreign body commonly enters – **right main bronchus**.
- Right lung lobes – **3**; left – **2**.
- Surfactant secreted by – **type II pneumocytes**; deficiency – **RDS in preterm**.
- Main muscle of inspiration – **diaphragm**; quiet expiration – **passive**.
- Tidal volume – **500 ml**; vital capacity – **~4.8 L**; residual volume – **~1.2 L**.
- Volume not measured by spirometry – **residual volume**.
- Anatomical dead space – **150 ml**.
- $O_2$ transported mostly as – **oxyhaemoglobin (~97 %)**; $CO_2$ mostly as – **bicarbonate (~70 %)**.
- Respiratory centre – **medulla oblongata**; pneumotaxic centre – **pons**.
- Main stimulus for respiration – **↑ $CO_2$ ($H^+$)**.
- Kussmaul breathing – **DKA (metabolic acidosis)**.
- Cheyne-Stokes – **heart failure, raised ICP, terminal stage**.
- Orthopnoea – **heart failure**.
- Cyanosis appears when deoxygenated Hb – **> 5 g/dl**.
- Normal $PaCO_2$ – **35–45 mmHg**; $PaO_2$ – **80–100 mmHg**.

# 7 Digestive System

## 7.1 Overview

- **Alimentary canal** (≈ **9 m**, mouth → anus) + **accessory organs** (teeth, tongue, salivary glands, liver, gall bladder, pancreas).
- **Processes** – ingestion, propulsion (**peristalsis**), mechanical digestion, chemical digestion (enzymes – **hydrolysis**), absorption, defaecation.
- **Wall layers (oesophagus → anus)** – **mucosa** (innermost), **submucosa (Meissner's plexus – secretion)**, **muscularis externa (inner circular + outer longitudinal; Auerbach's/ myenteric plexus – motility)**, **serosa/ adventitia**; **stomach has extra oblique layer**.
- **Peritoneum** – largest serous membrane; **greater omentum ("policeman of abdomen" – walls off infection)**, lesser omentum, mesentery; **retroperitoneal organs – kidneys, pancreas (mostly), duodenum (most), ascending & descending colon, aorta, IVC**.

### Digestive System – Organs & Functions




#### Digestion at a glance

**Mouth** – salivary amylase (ptyalin) – starch

**Stomach** – HCl (parietal cells), pepsin (chief cells), intrinsic factor → B12

**Liver** – bile (no enzyme) emulsifies fat

**Pancreas** – trypsin, lipase, amylase + bicarbonate; islets – insulin, glucagon

**Small intestine** – duodenum, jejunum, ileum; villi – most absorption
B12 & bile salts – terminal ileum

**Large intestine** – water, electrolytes, vitamin K by gut bacteria

**End products**

Carbohydrate → glucose

Protein → amino acids

Fat → fatty acids + glycerol

Iron absorbed – duodenum

Gastric emptying ≈ 2–4 hrs

Ampulla of Vater – 2nd part of duodenum

*Digestive organs & digestion at a glance*

## 7.2 Mouth, Teeth & Salivary Glands

- **Teeth** – **deciduous (milk) – 20** (erupt **6 months – 2.5 yrs**; first – **lower central incisor ~6 months**); **permanent – 32** (first – **first molar ~6 yrs**; last – **third molar/ wisdom 17–25 yrs**); **dental formula – permanent 2-1-2-3 (incisor, canine, premolar, molar) per quadrant**; deciduous **2-1-0-2** (no premolars).
- **Tooth structure** – **enamel (hardest substance in body)**, dentine, pulp (nerves, vessels), cementum; **fluoride** strengthens enamel (excess – fluorosis).

- **Tongue** – skeletal muscle; **taste buds on papillae** (circumvallate – largest, V-shaped, back; fungiform; foliate; filiform – no taste buds); **5 tastes – sweet, salty, sour, bitter, umami**; **taste – facial nerve (anterior ⅔ – chorda tympani), glossopharyngeal (posterior ⅓)**; **motor – hypoglossal (XII)**; general sensation anterior ⅔ – lingual nerve (V3).

| Salivary gland | Duct | Secretion | Point |
|---|---|---|---|
| **Parotid (largest)** | **Stensen's duct – opens opposite upper 2nd molar** | **Serous (watery, amylase)** | **Mumps**; facial nerve passes through |
| **Submandibular** | **Wharton's duct** – floor of mouth | Mixed – **maximum (~70 %) of saliva** | **Commonest site of salivary stones** |
| **Sublingual (smallest)** | Ducts of Rivinus | Mainly mucous | – |

- **Saliva** – **1–1.5 L/day**, pH **~6.8**; contains **salivary amylase (ptyalin – starch → maltose)**, lingual lipase, **lysozyme (antibacterial)**, IgA, mucus; secretion controlled by **ANS (parasympathetic – profuse watery)**.
- **Swallowing (deglutition)** – **oral (voluntary) → pharyngeal (involuntary – soft palate closes nasopharynx, epiglottis closes larynx, breathing stops) → oesophageal (peristalsis)**; **centre – medulla**.

## 7.3 Oesophagus & Stomach

- **Oesophagus** – **~25 cm**, C6 → T11; **upper ⅓ skeletal, middle mixed, lower ⅓ smooth muscle**; **4 constrictions** (at **15 cm – cricopharynx, narrowest**; 22.5 cm – aortic arch; 27.5 cm – left bronchus; **40 cm from incisors – diaphragm/ cardia**); **lower oesophageal sphincter** prevents reflux (GERD if weak); NG tube length – **nose-ear-xiphisternum (NEX)**.
- **Stomach** – J-shaped; **left hypochondrium & epigastrium**; parts – **cardia, fundus, body, antrum, pylorus (pyloric sphincter)**; **capacity ~1–1.5 L (up to 2–4 L)**; **rugae**; newborn capacity ~ **20–30 ml**.

| Gastric cell | Secretion |
|---|---|
| **Parietal (oxyntic)** | **HCl** (pH 1.5–3.5 – kills microbes, activates pepsin) & **intrinsic factor (vitamin B12 absorption – deficiency → pernicious anaemia)** |
| **Chief (peptic/ zymogen)** | **Pepsinogen → pepsin (protein → peptides)**, gastric lipase; **rennin (infants – curdles milk)** |
| **Mucous (neck) cells** | **Mucus + bicarbonate** – protect mucosa |
| **G cells (antrum)** | **Gastrin** – ↑ HCl & motility |
| **D cells** | Somatostatin (inhibits) |

- **Gastric juice ≈ 2–3 L/day**; **phases of secretion – cephalic (sight/ smell – vagus, ~30 %), gastric (food in stomach – most ~60 %), intestinal**.
- **Gastric emptying ≈ 2–4 hrs** – **fastest carbohydrates, slowest fats**; **chyme**; vomiting centre – **medulla (CTZ – area postrema)**.
- **Absorption in stomach** – minimal: **alcohol, water, aspirin**, some drugs.

## 7.4 Small Intestine

- **Length ≈ 6–7 m (longest part of GI tract)**; **duodenum (~25 cm, C-shaped, shortest & widest; 2nd part – ampulla of Vater/ major duodenal papilla – common bile duct + pancreatic duct, with sphincter of Oddi)**, **jejunum (~2.5 m)**, **ileum (~3.5 m – Peyer's patches; ends at ileocaecal valve)**.
- **Surface area ↑ by – circular folds (plicae circulares), villi (with lacteals & capillaries), microvilli (brush border) → ~200–250 m²**.
- **Succus entericus (intestinal juice)** – ~1.5–3 L/day; **brush-border enzymes – maltase, sucrase, lactase (deficiency – lactose intolerance)**, peptidases, enterokinase (activates trypsinogen).
- **Hormones – secretin (S cells – stimulates bicarbonate-rich pancreatic juice; first hormone discovered – Bayliss & Starling 1902)**, **cholecystokinin (CCK – I cells – gall bladder contraction & pancreatic enzyme secretion)**, GIP (↑ insulin), motilin (motility).
- **Absorption sites – most nutrients – jejunum**; **iron, calcium, folate – duodenum (& upper jejunum)**; **vitamin B12 (with IF) & bile salts – terminal ileum**; **fats – lacteals → lymph (chylomicrons)**; **glucose & galactose – $Na^+$-glucose co-transport (SGLT-1 – basis of ORS)**; fructose – facilitated diffusion.
- **Movements – segmentation (mixing)**, **peristalsis (propulsion)**; **migrating motor complex**.

## 7.5 Large Intestine

- **Length ≈ 1.5 m**; **caecum (with vermiform appendix – RIF; McBurney's point – ⅓ from ASIS to umbilicus) → ascending → transverse → descending → sigmoid colon → rectum (~12–15 cm) → anal canal (~4 cm)**; **taeniae coli (3 bands), haustra, appendices epiploicae**.
- **Functions – absorption of water (~1–1.5 L/day) & electrolytes ($Na^+$)**, **synthesis of vitamin K & some B vitamins (biotin) by gut flora**, formation & storage of faeces, mucus secretion.
- **Faeces** – ~100–200 g/day; **brown colour – stercobilin**; odour – indole, skatole; ~75 % water.
- **Defaecation reflex** – rectal distension → **internal anal sphincter (involuntary, smooth)** relaxes; **external anal sphincter (voluntary, skeletal – pudendal nerve)**; **gastrocolic reflex** – urge after meals (bowel training).
- **Anal canal – pectinate (dentate) line** – above: visceral, painless (internal haemorrhoids); below: somatic, painful (external haemorrhoids).

## 7.6 Liver & Biliary System

- **Liver – largest internal organ & largest gland (~1.2–1.5 kg)**; **right hypochondrium, epigastrium**; **4 lobes (right – largest, left, caudate, quadrate)**; **Glisson's capsule**; **functional unit – hepatic lobule (hexagonal; central vein, portal triad – hepatic artery, portal vein, bile duct)**; **Kupffer cells** – macrophages.
- **Dual blood supply – portal vein (~75 % – nutrient-rich from gut)** + **hepatic artery (~25 % – oxygenated)**; drains via **hepatic veins → IVC**; receives **~25 % of cardiac output**.
- **Functions – bile production (~0.5–1 L/day)**; **carbohydrate metabolism (glycogenesis, glycogenolysis, gluconeogenesis)**; **protein – albumin, clotting factors (I, II, V, VII, IX, X), deamination & urea formation (ornithine cycle)**; **fat – cholesterol, lipoproteins, β-oxidation, ketone bodies**; **storage – glycogen, vitamins A, D, B12, K, iron (ferritin), copper**; **detoxification & drug metabolism**, hormone inactivation, **bilirubin conjugation**, **heat production**, fetal haemopoiesis, Kupffer cell phagocytosis.
- **Regenerates** even after 70 % resection.

- **Bile** – **no digestive enzyme**; contains **bile salts (emulsify fats, help absorption of fat & fat-soluble vitamins A, D, E, K)**, bile pigments (bilirubin), cholesterol, lecithin; **enterohepatic circulation** (~95 % bile salts reabsorbed in terminal ileum).
- **Gall bladder** – pear-shaped, **stores & concentrates bile (~30–50 ml capacity)**; **CCK** → contraction; **ducts – right & left hepatic → common hepatic + cystic → common bile duct (CBD) → ampulla of Vater**; **Murphy's sign** (cholecystitis); **Calot's triangle**; gallstones – "**fat, female, fertile, forty, fair**".

## 7.7 Pancreas

- **Mixed gland** – retroperitoneal, behind stomach; **head (in C-loop of duodenum), neck, body, tail (touches spleen)**; ~ 15 cm.
- **Exocrine (~98–99 %) – acini** – **pancreatic juice ~1.2–1.5 L/day, alkaline (pH ~8, rich in $HCO_3^-$)** – **trypsinogen (→ trypsin by enterokinase), chymotrypsinogen, carboxypeptidase**, **pancreatic amylase**, **pancreatic lipase (main fat-digesting enzyme)**, nucleases; **duct of Wirsung (main) & Santorini (accessory)**.
- **Endocrine (~1–2 %) – islets of Langerhans (~1 million)** – **α cells – glucagon (↑ glucose)**, **β cells (~70 %) – insulin (↓ glucose)**, **δ cells – somatostatin**, PP/F cells – pancreatic polypeptide.
- **Serum amylase & lipase ↑ – acute pancreatitis** (lipase more specific).

## 7.8 Digestion Summary & Metabolism

| Nutrient | Enzyme (site) | End product |
|---|---|---|
| **Carbohydrates** | Salivary amylase (mouth) → pancreatic amylase → maltase, sucrase, lactase (brush border) | **Monosaccharides – glucose, fructose, galactose** |
| **Proteins** | **Pepsin (stomach)** → trypsin, chymotrypsin, carboxypeptidase (pancreas) → peptidases | **Amino acids** |
| **Fats** | Bile emulsifies → **pancreatic lipase** (main) | **Fatty acids + glycerol (monoglycerides)** |
| **Nucleic acids** | Nucleases | Nucleotides → bases, sugars |

- **Energy value** – **carbohydrate 4 kcal/g, protein 4 kcal/g, fat 9 kcal/g**, alcohol 7 kcal/g.
- **BMR** – energy at complete rest (~1 kcal/kg/hr; ↑ by thyroxine, fever – 7 % per °F/ ~13 % per °C, pregnancy, children; ↓ in sleep, starvation, hypothyroidism); **highest BMR per kg – infants**.
- **Glycolysis** – glucose → pyruvate (cytoplasm, 2 ATP net); **Krebs (TCA) cycle** – mitochondria; **ETC** – most ATP (~30–32 ATP/glucose total).
- **Ketone bodies** (acetoacetate, β-hydroxybutyrate, acetone) – in **starvation, uncontrolled DM** (fat breakdown).
- **Normal fasting blood glucose 70–100 mg/dl**; post-prandial < 140; **storage form – glycogen (liver & muscle)**.

**♫ Yaad karne ki trick**

- Small intestine parts – **"Dow Jones Industrial"** – Duodenum, Jejunum, Ileum.
- Absorption – **"Iron in Duo, B12 at the End (terminal ileum)"**.
- Large intestine water absorption – **"Colon = Collect water"**.
- Gallstones – **"5 F"** – fat, female, fertile, forty, fair.

### PYQ style

- Largest gland – **liver**; largest salivary gland – **parotid**.
- Hardest substance – **enamel**; deciduous teeth – **20**; permanent – **32**.
- Salivary enzyme – **ptyalin (amylase)**.
- Parotid duct – **Stensen's**; submandibular – **Wharton's**.
- Swallowing centre – **medulla**; oesophagus length – **25 cm**.
- HCl & intrinsic factor – **parietal cells**; pepsinogen – **chief cells**.
- Intrinsic factor deficiency – **pernicious anaemia**.
- Rennin – **curdles milk (infants)**.
- Longest part of GI tract – **small intestine**; most absorption – **jejunum**.
- B12 absorbed in – **terminal ileum**; iron in – **duodenum**.
- Villi lymph vessel – **lacteal (fat absorption)**.
- First hormone discovered – **secretin**.
- Gall bladder contraction – **CCK**.
- Bile is produced by – **liver**; stored in – **gall bladder**; contains – **no enzyme**.
- Vitamin K synthesised by – **gut bacteria (large intestine)**.
- Main fat-digesting enzyme – **pancreatic lipase**.
- Trypsinogen activated by – **enterokinase**.
- Insulin – **β cells**; glucagon – **α cells**.
- Liver's functional unit – **hepatic lobule**; macrophages – **Kupffer cells**.
- Urea is formed in – **liver**.
- Appendix – **right iliac fossa (McBurney's point)**.
- Fat gives – **9 kcal/g**; protein & carbs – **4 kcal/g**.

# 8 Urinary System, Body Fluids, Electrolytes & Acid–Base

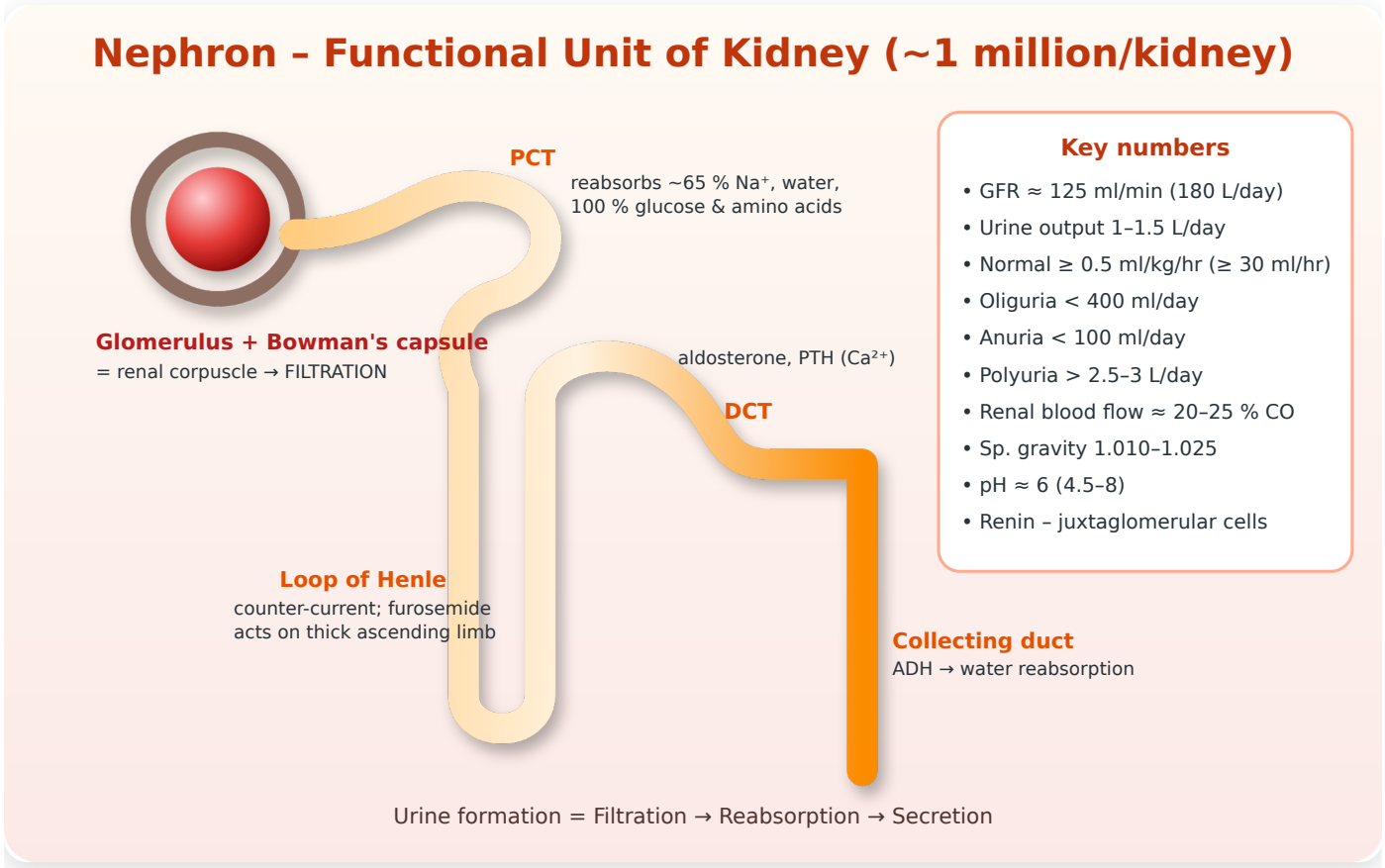
## 8.1 Kidneys

- **Location** – **retroperitoneal, T12–L3**; **right kidney slightly lower (liver above)**; bean-shaped, **~11–12 cm × 6 × 3 cm, ~130–150 g**; **hilum** – renal artery, vein, ureter (**anterior to posterior – vein, artery, pelvis "VAP"**).
- **Coverings** – fibrous capsule, perirenal fat, renal fascia (Gerota's).
- **Internal** – **cortex (outer – glomeruli)**, **medulla (renal pyramids 8–18 – loops & collecting ducts)**, **renal papilla → minor calyx → major calyx → renal pelvis → ureter**; renal columns.
- **Blood supply** – **renal artery (from aorta, L1–L2)**; receives **~20–25 % of cardiac output (~1.2 L/min)**; afferent arteriole → **glomerulus** → **efferent arteriole** → peritubular capillaries/ **vasa recta**.

### Functions of kidney

- **Excretion of waste** – **urea, uric acid, creatinine**, drugs, toxins.
- **Water & electrolyte balance**, **acid–base balance (excretes $H^+$, reabsorbs $HCO_3^-$)**, **BP regulation (renin – RAAS)**.
- **Endocrine** – **erythropoietin (RBC production – CKD → anaemia)**, **calcitriol (active vitamin D – 1α-hydroxylation; CKD → renal osteodystrophy)**, **renin**, prostaglandins.
- **Gluconeogenesis** (prolonged fasting).

## 8.2 Nephron & Urine Formation




*Nephron – parts & functions with key renal numbers*

- **Nephron – structural & functional unit (~1 million per kidney)**; **cortical (85 %)** & **juxtamedullary (15 % – long loops, concentrate urine)**.
- **Parts – renal corpuscle (glomerulus + Bowman's capsule) → PCT → loop of Henle (descending – water permeable; thick ascending – impermeable to water, $Na^+$-$K^+$-$2Cl^-$ pump) → DCT → collecting duct**.
- **Juxtaglomerular apparatus – JG cells (afferent arteriole – secrete renin) + macula densa (DCT – senses NaCl)** + extraglomerular mesangial cells.

| Step | Site | Details |
|---|---|---|
| **1. Glomerular filtration** | Glomerulus | **Pressure-driven ultrafiltration**; **GFR ≈ 125 ml/min (180 L/day)**; filtrate = plasma minus proteins & cells; **net filtration pressure ≈ 10 mmHg** (glomerular hydrostatic 55 − oncotic 30 − capsular 15) |
| **2. Tubular reabsorption** | **PCT (maximum ~65–70 %)** | **100 % glucose & amino acids**, $Na^+$, water, $HCO_3^-$, $K^+$; **renal threshold for glucose ≈ 180 mg/dl** (glycosuria above) |
| | Loop of Henle | **Counter-current multiplier** → concentrated medulla |
| | DCT | **Aldosterone – $Na^+$ reabsorption, $K^+$ & $H^+$ secretion**; **PTH – $Ca^{2+}$ reabsorption** |
| | Collecting duct | **ADH (vasopressin) – aquaporins → water reabsorption (concentrated urine)**; ADH deficiency → diabetes insipidus |
| **3. Tubular secretion** | PCT, DCT | **$H^+$, $K^+$, $NH_4^+$, creatinine, drugs (penicillin, PAH)** |

- **~99 % of filtrate reabsorbed** → urine ~1–1.5 L/day.
- **Clearance** – **inulin clearance = true GFR**; **creatinine clearance – clinical GFR estimate (~100–125 ml/min)**; **PAH clearance – renal plasma flow (~625 ml/min)**.
- **Hormones acting** – **ADH (posterior pituitary – water)**, **aldosterone (adrenal cortex – $Na^+$ in, $K^+$ out)**, **ANP (atria – ↑ $Na^+$ & water excretion)**, **PTH**.

## 8.3 Ureters, Bladder & Urethra

- **Ureters** – **~25–30 cm**, **peristalsis** propels urine; **3 constrictions** – **pelvi-ureteric junction**, **crossing iliac vessels (pelvic brim)**, **vesico-ureteric junction (narrowest – commonest site of stone impaction)**; transitional epithelium.
- **Urinary bladder** – **hollow muscular organ (detrusor muscle)** in pelvis behind pubic symphysis; **capacity ~500–600 ml (max ~800–1000)**; **urge to void at ~200–300 ml (first desire ~150–200)**; **trigone** – smooth triangular area between 2 ureteric openings & internal urethral orifice (no rugae; infection-prone); lined with **transitional epithelium**.
- **Urethra** – **female ~4 cm** (short → **UTI more common**; catheter insert **5–7.5 cm (2–3 inches)**); **male ~18–20 cm** (prostatic, membranous – narrowest/ least dilatable part except external meatus, spongy/ penile; catheter insert **17–22 cm (7–9 inches)**).
- **Sphincters** – **internal (smooth, involuntary)**, **external (skeletal, voluntary – pudendal nerve)**.
- **Micturition reflex** – bladder stretch → **sacral spinal cord S2–S4 (parasympathetic – pelvic splanchnic nerves) → detrusor contraction & internal sphincter relaxation**; **voluntary control – pons (pontine micturition centre) & cortex**; **sympathetic (T11–L2) – storage**.
- **Bladder control** develops ~2–3 yrs (day), night ~ by 5 yrs (enuresis beyond).

**Normal urine**

| Parameter | Normal |
|---|---|
| **Volume** | **1–1.5 L/day (≥ 0.5 ml/kg/hr; ≥ 30 ml/hr adult)**; child **1–2 ml/kg/hr** |
| Colour | **Pale yellow/ amber (urochrome)** |
| **Specific gravity** | **1.010–1.025** (1.003–1.030 range) |
| **pH** | **~6 (4.5–8)** – slightly acidic |
| Odour | Aromatic; ammoniacal on standing; **fruity (ketones)**; mousy (PKU) |
| Abnormal constituents | **Protein, glucose, ketones, blood, bile, pus cells** (normally absent or trace) |

- **Oliguria < 400 ml/day (< 0.5 ml/kg/hr)**, **anuria < 100 ml/day**, **polyuria > 2.5–3 L/day**, **nocturia, dysuria, frequency, urgency, haematuria, pyuria, enuresis, retention, incontinence**.
- **Blood urea 15–40 mg/dl (BUN 7–20)**, **serum creatinine 0.6–1.2 mg/dl**, **uric acid 3–7 mg/dl**.

## 8.4 Body Fluids

- **Total body water (TBW)** – **~60 % of body weight in adult male (~42 L in 70 kg)**, **~50–55 % female** (more fat), **~75–80 % newborn** (higher risk of dehydration), ~50 % elderly.
- **Intracellular fluid (ICF) – ⅔ (~40 % body weight; 28 L)**; **extracellular fluid (ECF) – ⅓ (~20 %; 14 L)** = **interstitial (¾ of ECF ~10.5 L)** + **plasma/ intravascular (¼ ~3.5 L)** + transcellular (CSF, synovial, pleural…).

- **Daily intake ≈ 2.5 L** (drinking ~1.5 L, food ~0.8 L, **metabolic water ~0.3 L**); **output ≈ 2.5 L** (**urine ~1.5 L**, skin/ sweat ~0.5 L, lungs ~0.3–0.4 L, faeces ~0.1–0.2 L); **insensible loss ≈ 800–1000 ml/day** (↑ 10–12 % per °C fever).
- **Maintenance fluid (Holliday-Segar)** – **100 ml/kg for first 10 kg + 50 ml/kg next 10 kg + 20 ml/kg each kg above 20** (per 24 hrs).
- **Osmolality** – **plasma 275–295 mOsm/kg**; **thirst centre – hypothalamus**.
- **Starling forces** – capillary hydrostatic pressure (out) vs plasma oncotic pressure (in – albumin); **oedema** – ↑ hydrostatic (heart failure), ↓ oncotic (hypoalbuminaemia – nephrotic, cirrhosis, malnutrition), ↑ permeability (inflammation, burns), lymphatic obstruction (filariasis).

## 8.5 Electrolytes – Normal Values & Key Facts

| Electrolyte | Normal | Key facts |
|---|---|---|
| **Sodium ($Na^+$)** | **135–145 mEq/L** | **Main ECF cation**; water balance, nerve impulse; hypo – confusion, seizures; hyper – thirst, CNS |
| **Potassium ($K^+$)** | **3.5–5.0 mEq/L** | **Main ICF cation**; **cardiac & muscle function**; hypo – weakness, **U waves**, arrhythmia; hyper – **tall T, arrhythmia, cardiac arrest** |
| **Chloride ($Cl^-$)** | **96–106 mEq/L** | Main ECF anion |
| **Bicarbonate ($HCO_3^-$)** | **22–26 mEq/L** | Main buffer |
| **Calcium (total)** | **8.5–10.5 mg/dl** (ionised 4.5–5.3 mg/dl) | Bones, clotting, muscle contraction; hypo – **tetany, Chvostek's & Trousseau's signs**, carpopedal spasm |
| **Magnesium** | **1.5–2.5 mEq/L** (1.7–2.4 mg/dl) | Second ICF cation; hypo → arrhythmia, tetany |
| **Phosphate** | **2.5–4.5 mg/dl** | Main ICF anion; inverse with calcium |
| **Anion gap** | **8–12 ($Na - (Cl + HCO_3)$)** | ↑ in DKA, lactic acidosis, renal failure, poisoning |

## 8.6 Acid–Base Balance

- **Normal arterial pH 7.35–7.45** (**acidosis < 7.35, alkalosis > 7.45**; **compatible with life ~6.8–7.8**); **$PaCO_2$ 35–45 mmHg** (respiratory component); **$HCO_3^-$ 22–26 mEq/L** (metabolic component); **$PaO_2$ 80–100 mmHg**; **$SaO_2$ 95–100 %**; **base excess −2 to +2**.
- **Henderson-Hasselbalch** – pH ∝ $HCO_3^-$ / $PaCO_2$ (ratio **20 : 1**).
- **Regulation** – **1. buffers (seconds – bicarbonate (most important ECF), phosphate (ICF & urine), proteins/ haemoglobin (most abundant intracellular))**, **2. lungs (minutes – $CO_2$ excretion)**, **3. kidneys (hours–days – most powerful; $H^+$ excretion, $HCO_3^-$ reabsorption, ammonia)**.

| Disorder | pH | Primary change | Compensation | Causes |
|---|---|---|---|---|
| **Respiratory acidosis** | ↓ | ↑ **$PaCO_2$** | ↑ $HCO_3^-$ (kidney) | **Hypoventilation – COPD, respiratory depression (opioids), neuromuscular weakness, airway obstruction** |
| **Respiratory alkalosis** | ↑ | ↓ **$PaCO_2$** | ↓ $HCO_3^-$ | **Hyperventilation – anxiety/ panic, high altitude, early salicylate poisoning, fever, PE, ventilator** |

| Disorder | pH | Primary change | Compensation | Causes |
|---|---|---|---|---|
| **Metabolic acidosis** | ↓ | **↓ $HCO_3^-$** | ↓ $PaCO_2$ (**Kussmaul**) | **DKA, lactic acidosis (shock), renal failure, diarrhoea ($HCO_3^-$ loss), methanol/ salicylate** |
| **Metabolic alkalosis** | ↑ | **↑ $HCO_3^-$** | ↑ $PaCO_2$ (hypoventilation) | **Vomiting/ NG suction (HCl loss), diuretics, hypokalaemia, antacid excess** |

- **ROME** – **R**espiratory **O**pposite (pH ↑ → $PaCO_2$ ↓), **M**etabolic **E**qual (pH & $HCO_3^-$ same direction).
- **ABG steps** – 1. pH (acidosis/ alkalosis) → 2. $PaCO_2$ (respiratory?) → 3. $HCO_3^-$ (metabolic?) → 4. compensation (partial if pH abnormal, full if pH normal) → 5. oxygenation.
- **Example** – pH 7.30, $PaCO_2$ 55, $HCO_3^-$ 26 → **uncompensated respiratory acidosis**; pH 7.48, $PaCO_2$ 40, $HCO_3^-$ 32 → **metabolic alkalosis**.
- **ABG sample** – **radial artery (Allen's test first)**, heparinised syringe, no air bubbles, **ice/ analyse within 15–30 min**, pressure 5 min (10 if anticoagulated).

**♫ Yaad karne ki trick**

- **ROME** – Respiratory Opposite, Metabolic Equal.
- Hilum structures (front → back) – **"VAP"** – Vein, Artery, Pelvis.
- Vomiting = **metabolic alkalosis** (acid loss); Diarrhoea = **metabolic acidosis** (base loss) – "**Top loses acid, bottom loses base**".

### PYQ style

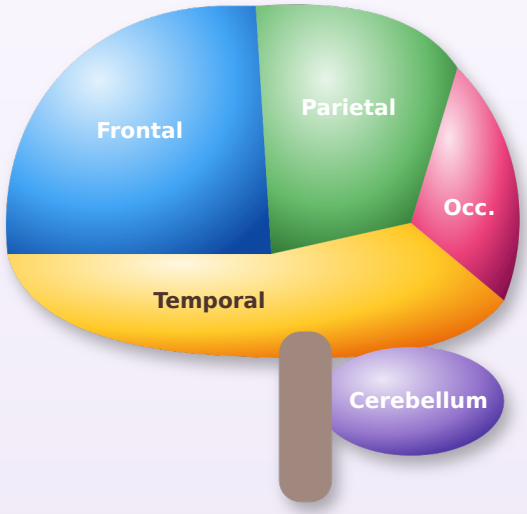
- Functional unit of kidney – **nephron (~1 million each)**.
- Kidney level – **T12–L3**; lower kidney – **right**.
- Renin secreted by – **juxtaglomerular cells**.
- Normal GFR – **125 ml/min (180 L/day)**.
- Maximum reabsorption – **PCT**; glucose reabsorbed – **100 % in PCT**.
- Renal threshold for glucose – **180 mg/dl**.
- ADH acts on – **collecting duct**; aldosterone on – **DCT ($Na^+$ in, $K^+$ out)**.
- Best measure of GFR – **inulin clearance**; clinically – **creatinine clearance**.
- Female urethra – **4 cm**; male – **18–20 cm**.
- Bladder capacity – **~500–600 ml**; urge at – **~200–300 ml**.
- Micturition centre (spinal) – **S2–S4**.
- Normal urine output – **≥ 30 ml/hr / 0.5 ml/kg/hr**; oliguria – **< 400 ml/day**.
- Normal specific gravity – **1.010–1.025**.
- Kidney hormone for RBC – **erythropoietin**.
- TBW in adult – **60 %**; ICF – **⅔**.
- Main ICF cation – **$K^+$**; ECF – **$Na^+$**.
- Normal serum $K^+$ – **3.5–5.0 mEq/L**; $Na^+$ – **135–145**.
- Tetany – **hypocalcaemia (Chvostek, Trousseau)**.
- Normal pH – **7.35–7.45**; $HCO_3^-$ – **22–26**; $PaCO_2$ – **35–45**.
- Vomiting – **metabolic alkalosis**; diarrhoea – **metabolic acidosis**.
- COPD/ opioid overdose – **respiratory acidosis**; anxiety hyperventilation – **respiratory alkalosis**.
- Most powerful acid–base regulator – **kidney**; fastest – **buffers**.
- Before radial ABG – **Allen's test**.

# 9 Nervous System

## 9.1 Organisation & Cells

- **CNS** – brain + spinal cord; **PNS** – **12 pairs cranial + 31 pairs spinal nerves**; PNS → **somatic (voluntary)** & **autonomic (involuntary – sympathetic & parasympathetic)**; **afferent (sensory)** & **efferent (motor)**.
- **Neuron – structural & functional unit**; parts – **cell body (soma, Nissl bodies)**, **dendrites (carry impulse towards cell body)**, **axon (away from cell body; axon hillock – impulse origin)**, **myelin sheath**, **nodes of Ranvier**, axon terminals; **neurons do not divide (no centrioles)** – poor regeneration in CNS.
- **Types** – **multipolar (motor – most common)**, **bipolar (retina, olfactory, inner ear)**, **pseudo-unipolar (sensory – dorsal root ganglion)**.
- **Neuroglia** (supporting – more numerous than neurons): **astrocytes (most abundant – blood-brain barrier)**, **oligodendrocytes (myelin in CNS)**, **microglia (phagocytes – macrophages of CNS)**, **ependymal cells (line ventricles – CSF)**; PNS – **Schwann cells (myelin in PNS – aid regeneration)**, satellite cells.
- **Grey matter** – cell bodies (cortex outside in brain; **H-shaped inside in spinal cord**); **white matter** – myelinated axons.

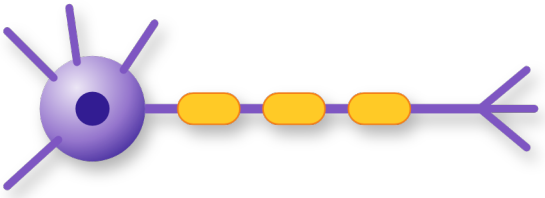
### Brain Lobes & Neuron




Brainstem (midbrain, pons, medulla)

Frontal – personality, motor (Broca's speech area)
Parietal – sensation · Occipital – vision
Temporal – hearing, memory (Wernicke's) · Cerebellum – balance
Medulla – vital centres (respiration, cardiac, vasomotor)

**Neuron (structural & functional unit)**



Dendrites → impulse TOWARDS cell body
Axon → impulse AWAY from cell body
Myelin sheath (Schwann cells – PNS, oligodendrocytes – CNS) → saltatory conduction via nodes of Ranvier
Synapse – neurotransmitter (ACh, etc.)
Resting potential ≈ −70 mV
Neurons don't divide (no centrioles)
CSF ≈ 150 ml, ~500 ml/day (choroid plexus)
12 pairs cranial · 31 pairs spinal nerves
Spinal cord ends at L1–L2 (adult)
Brain weight ≈ 1.3–1.4 kg

*Brain lobes with functions & structure of a neuron*

### Nerve impulse

- **Resting membrane potential ≈ −70 mV** ($Na^+$-$K^+$ pump, $K^+$ leak); **depolarisation – $Na^+$ influx** (to ~+30 mV); **repolarisation – $K^+$ efflux**; hyperpolarisation; **threshold ≈ −55 mV**; **all-or-none law**;

**refractory period**.

- **Saltatory conduction** – impulse jumps node to node in myelinated fibres (**faster – up to ~120 m/s**); demyelination (multiple sclerosis, GBS) slows conduction.
- **Synapse** – presynaptic terminal releases **neurotransmitter ($Ca^{2+}$-dependent exocytosis)** → synaptic cleft → postsynaptic receptors; **one-way conduction**; **synaptic delay ~0.5 ms**.

| Neurotransmitter | Key role / disorder |
|---|---|
| **Acetylcholine** | NMJ, parasympathetic, memory (**↓ in Alzheimer's**); myasthenia (receptor antibodies) |
| **Dopamine** | Movement (**↓ in Parkinson's – substantia nigra**), reward; **↑ in schizophrenia** |
| **Noradrenaline** | Sympathetic, mood (↓ depression) |
| **Serotonin (5-HT)** | Mood, sleep, appetite (**↓ depression**) |
| **GABA** | **Main inhibitory (brain)** – BZD act here; ↓ in seizures, anxiety |
| **Glycine** | Inhibitory (spinal cord) – **tetanus toxin & strychnine block** |
| **Glutamate** | **Main excitatory** |
| **Endorphins/ enkephalins** | Natural analgesics |

## 9.2 Brain

- **Weight ≈ 1.3–1.4 kg (~2 % body weight)**; uses **~20 % of body $O_2$ & ~15 % of cardiac output (~750 ml/min)**; **glucose – main fuel**; **irreversible damage after ~4–6 min of anoxia**.
- **Parts** – **forebrain (cerebrum + diencephalon)**, **midbrain**, **hindbrain (pons, medulla, cerebellum)**; **brainstem = midbrain + pons + medulla**.

### Cerebrum (largest part)

- **2 hemispheres** joined by **corpus callosum (largest commissure)**; **gyri & sulci**; **central sulcus (of Rolando)** separates frontal & parietal; **lateral sulcus (Sylvian fissure)** separates temporal; **longitudinal fissure** between hemispheres; **left hemisphere dominant in ~95 % (language)**.

| Lobe | Functions |
|---|---|
| **Frontal (largest)** | **Primary motor cortex (precentral gyrus – motor homunculus)**, **Broca's area (motor speech – left inferior frontal; lesion → expressive/ non-fluent aphasia)**, personality, judgment, intellect, planning, emotions |
| **Parietal** | **Primary somatosensory cortex (postcentral gyrus)** – touch, pain, temperature, proprioception; spatial orientation |
| **Temporal** | **Auditory cortex (hearing)**, **Wernicke's area (understanding speech – lesion → receptive/ fluent aphasia)**, **memory (hippocampus)**, smell |
| **Occipital (smallest)** | **Visual cortex** – vision |
| **Limbic system** | **Emotions, behaviour, memory** – hippocampus (memory), amygdala (fear, aggression) |
| **Insula** | Taste, visceral sensation |

- **Basal ganglia/ nuclei** – caudate, putamen, globus pallidus, substantia nigra – **control of movement** (Parkinson's, Huntington's chorea).

- **Motor pathways cross (decussate) – right brain controls left body** (pyramidal decussation in medulla).

### Diencephalon

- **Thalamus** – **relay centre for all sensations (except smell)**.
- **Hypothalamus** – **"master of autonomic & endocrine control"** – **temperature regulation (thermostat)**, **hunger & satiety**, **thirst & water balance (ADH, oxytocin synthesised in supraoptic & paraventricular nuclei)**, sleep–wake (circadian – suprachiasmatic nucleus), emotions, controls **pituitary (releasing hormones)**, sexual behaviour.
- **Epithalamus** – **pineal gland (melatonin – sleep; calcified – landmark on X-ray)**.

### Brainstem & cerebellum

- **Midbrain** – visual & auditory reflexes (superior & inferior colliculi), **substantia nigra (dopamine)**, cerebral aqueduct, CN III & IV nuclei.
- **Pons** – bridge; **pneumotaxic & apneustic respiratory centres**, CN V–VIII nuclei; pontine micturition centre.
- **Medulla oblongata** – **vital centres – cardiac, respiratory, vasomotor**; reflex centres – **vomiting, coughing, sneezing, swallowing**; **pyramidal decussation**; CN IX–XII; **coning (tonsillar herniation through foramen magnum) → death**.
- **Reticular activating system (RAS)** – brainstem – **consciousness, wakefulness, arousal**.
- **Cerebellum ("little brain", 2nd largest)** – **balance (equilibrium), posture, coordination of voluntary movements, muscle tone**; lesion → **ataxia, intention tremor, nystagmus, dysdiadochokinesia** (same side); alcohol affects.

## 9.3 Protection – Meninges, CSF & BBB

- **Meninges (outer → inner)** – **Dura mater (tough, double layer – dural venous sinuses)**, **arachnoid mater**, **pia mater (adherent to brain)** – "**DAP – Dura, Arachnoid, Pia**".
- **Spaces** – **epidural (extradural – middle meningeal artery bleed → extradural haematoma, lucid interval; epidural anaesthesia in spine)**, **subdural (bridging veins → subdural haematoma – elderly, alcoholics)**, **subarachnoid (CSF – subarachnoid haemorrhage – berry aneurysm, "worst headache of life"; spinal anaesthesia & LP)**.
- **Ventricles** – **2 lateral → (foramen of Monro) → 3rd → (cerebral aqueduct of Sylvius – commonest site of obstruction → hydrocephalus) → 4th → (foramina of Magendie & Luschka) → subarachnoid space**.
- **CSF** – **formed by choroid plexus (mainly lateral ventricles)**; **~500 ml/day; total volume ~120–150 ml**; **absorbed by arachnoid villi/ granulations into dural venous sinuses (superior sagittal)**; **clear, colourless; pressure 60–180 mm$H_2O$ (lying)**; **protein 15–45 mg/dl, glucose 50–80 mg/dl (~⅔ of blood glucose)**, cells 0–5 lymphocytes/µl; functions – cushion/ shock absorber, buoyancy, nutrition, waste removal.
- **CSF in meningitis** – **bacterial: turbid, ↑ neutrophils, ↑ protein, ↓↓ glucose**; **viral: clear, lymphocytes, normal glucose**; **TB: cobweb clot, lymphocytes, ↓ glucose, ↑↑ protein**.
- **Blood-brain barrier** – capillary endothelium tight junctions + astrocytes – blocks most drugs, toxins; lipid-soluble substances cross; immature in neonates (**kernicterus**).
- **Intracranial pressure (ICP)** – **normal 5–15 mmHg** (0–10 in some texts); **Monro-Kellie doctrine** (brain + blood + CSF = constant); **Cushing's triad – hypertension (widened pulse pressure),**

**bradycardia, irregular respiration** – late sign of ↑ ICP.

## 9.4 Spinal Cord & Reflexes

- **Spinal cord** – from **foramen magnum (medulla) to L1–L2 in adults (conus medullaris)**; **L3 in newborn**; **~45 cm (male), 42–43 cm (female)**; **cervical & lumbar enlargements** (limb plexuses); **cauda equina** – nerve roots below L1/2; **filum terminale**.
- **Lumbar puncture at L3–L4 or L4–L5** – below cord (safe); position – **lateral (fetal) or sitting, flexed**; post-LP – lie flat (per protocol), fluids.
- **Cross-section** – **H-shaped grey matter** – **anterior (ventral) horn – motor**, **posterior (dorsal) horn – sensory**, lateral horn (T1–L2 – sympathetic); **dorsal root ganglion – sensory cell bodies**; **dorsal root – sensory, ventral root – motor** ("**DAVE** – Dorsal Afferent, Ventral Efferent"); **spinal nerve – mixed**.

| Tract | Carries |
|---|---|
| **Dorsal column (fasciculus gracilis & cuneatus)** | **Fine touch, vibration, proprioception (position sense)** – cross in medulla |
| **Lateral spinothalamic** | **Pain & temperature** – cross in spinal cord (1–2 segments up) |
| **Anterior spinothalamic** | Crude touch, pressure |
| **Spinocerebellar** | Unconscious proprioception |
| **Corticospinal (pyramidal) – lateral** | **Voluntary motor** – crosses in medulla; **UMN lesion → spasticity, hyperreflexia, Babinski +ve**; **LMN lesion → flaccidity, hyporeflexia, atrophy, fasciculations** |

- **Reflex arc** – **receptor → sensory (afferent) neuron → integration centre (spinal cord) → motor (efferent) neuron → effector**; **knee jerk (patellar) – monosynaptic stretch reflex – L2–L4 (mainly L3, L4)**; **ankle jerk – S1–S2**; **biceps – C5–C6**; **triceps – C7**; **supinator – C5–C6**; withdrawal reflex – polysynaptic.
- **Babinski sign (extensor plantar)** – **UMN lesion**; **normal up to ~1–2 yrs** (immature pyramidal tract).
- **Grading reflexes** – 0 absent, 1+ diminished, **2+ normal**, 3+ brisk, 4+ clonus.

## 9.5 Cranial Nerves (12 pairs)

| No. | Name | Type | Function / test |
|---|---|---|---|
| **I** | **Olfactory** | S | **Smell** (test each nostril – coffee, soap) |
| **II** | **Optic** | S | **Vision** – acuity (Snellen), fields, fundus |
| **III** | **Oculomotor** | M | **Most eye movements, eyelid elevation, pupil constriction (parasympathetic), accommodation**; palsy → **ptosis, eye "down & out", dilated pupil** |
| **IV** | **Trochlear (smallest, thinnest, longest intracranial course)** | M | **Superior oblique – eye down & in** |
| **V** | **Trigeminal (largest)** | B | **Facial sensation (3 divisions – ophthalmic, maxillary, mandibular)**, **muscles of mastication**, **corneal reflex (afferent)**; **trigeminal neuralgia** |
| **VI** | **Abducens** | M | **Lateral rectus – abduction**; palsy → convergent squint (↑ ICP – long course over petrous apex → **false localising sign**) |

| No. | Name | Type | Function / test |
|---|---|---|---|
| **VII** | **Facial** | B | **Facial expression**, **taste anterior ⅔ tongue**, lacrimal & salivary (submandibular, sublingual) secretion, corneal reflex (efferent – eye closure); **Bell's palsy** |
| **VIII** | **Vestibulocochlear (auditory)** | S | **Hearing (Rinne, Weber) & balance** |
| **IX** | **Glossopharyngeal** | B | **Taste posterior ⅓**, **gag reflex (afferent)**, parotid secretion, swallowing; carotid sinus |
| **X** | **Vagus (longest, most widely distributed)** | B | **Parasympathetic to heart, lungs, GI tract (up to ⅔ transverse colon)**, swallowing, **voice (recurrent laryngeal)**, **gag reflex (efferent)**; "say ah" – uvula deviation |
| **XI** | **Accessory (spinal accessory)** | M | **Sternocleidomastoid & trapezius – shoulder shrug, head turning** |
| **XII** | **Hypoglossal** | M | **Tongue movements** – lesion → tongue deviates to **same side** |

- **Parasympathetic fibres** in **III, VII, IX, X** ("**1973**" → 10, 9, 7, 3).
- **Mnemonic names** – "**Oh Oh Oh To Touch And Feel Very Good Velvet, Ahh Heaven**"; type – "**Some Say Marry Money, But My Brother Says Big Brains Matter More**" (S = sensory, M = motor, B = both).

## 9.6 Spinal Nerves & Plexuses

- **31 pairs** – **cervical 8, thoracic 12, lumbar 5, sacral 5, coccygeal 1** (**C8 exists though only 7 cervical vertebrae**).
- **Dermatome** – skin area supplied by one spinal nerve (**C6 – thumb, T4 – nipple, T10 – umbilicus, L1 – inguinal, L4 – knee/ medial leg, S1 – lateral foot/ little toe**); herpes zoster follows dermatomes.

| Plexus | Roots | Key nerves |
|---|---|---|
| **Cervical** | C1–C4 | **Phrenic (C3–C5 – diaphragm)** |
| **Brachial** | **C5–T1** | **Musculocutaneous, axillary (deltoid – surgical neck humerus #), radial (wrist drop – crutch palsy, shaft of humerus #), median (carpal tunnel syndrome, "ape hand"; "labourer's nerve"), ulnar (claw hand; "funny bone" – medial epicondyle; "musician's nerve")**; **Erb's palsy (C5–C6 – "waiter's tip")**, **Klumpke's (C8–T1 – claw hand)** – birth injuries |
| **Lumbar** | L1–L4 | **Femoral (quadriceps – knee extension)**, obturator (adductors) |
| **Sacral** | L4–S4 | **Sciatic (L4–S3) – largest & longest nerve in body** → tibial & **common peroneal (foot drop – fibular neck)**; **pudendal (S2–S4 – external sphincters, perineum; pudendal block)** |

## 9.7 Autonomic Nervous System

| Feature | Sympathetic ("fight or flight") | Parasympathetic ("rest & digest") |
|---|---|---|
| Origin | **Thoracolumbar (T1–L2)** | **Craniosacral (CN III, VII, IX, X + S2–S4)** |
| Ganglia | Near spinal cord (sympathetic chain) – **short preganglionic, long postganglionic** | Near/ in organ – **long preganglionic, short postganglionic** |
| Neurotransmitter (post-ganglionic) | **Noradrenaline** (except sweat glands – ACh) | **Acetylcholine** |

| Feature | Sympathetic ("fight or flight") | Parasympathetic ("rest & digest") |
|---|---|---|
| Pupil | **Dilation (mydriasis)** | **Constriction (miosis)** |
| Heart | **↑ rate & force** | **↓ rate** (vagus) |
| Bronchi | **Dilation** | Constriction, ↑ secretion |
| GI | **↓ motility & secretion**, sphincters contract | **↑ motility & secretion** |
| Bladder | Detrusor relaxes, sphincter contracts (**retention**) | **Detrusor contracts (micturition)** |
| Blood glucose | **↑ (glycogenolysis)** | – |
| Salivary | Thick, scanty | **Profuse watery** |
| Sex | **Ejaculation** | **Erection** ("**P**oint & **S**hoot") |
| Adrenal medulla | **Adrenaline release** (modified sympathetic ganglion) | – |

- **Preganglionic neurotransmitter – ACh in both (nicotinic receptors)**.

## 9.8 Consciousness, Sleep & GCS

- **Sleep stages** – **NREM (stages N1, N2, N3 – deep/ slow-wave sleep: GH release, night terrors, sleepwalking, enuresis)** and **REM (dreaming, rapid eye movements, muscle atonia, nightmares; ~20–25 % in adults; ~50 % in newborn)**; cycle **~90 min**; newborn sleeps **16–18 hrs/day**, adult 7–8 hrs.
- **EEG waves** – **alpha (awake relaxed, eyes closed)**, beta (alert), theta (children/ drowsy), **delta (deep sleep)**.
- **Glasgow Coma Scale (Teasdale & Jennett, 1974)** – **Eye opening (4) + Verbal (5) + Motor (6) = 3–15**; **≤ 8 – coma/ severe head injury (intubate – "8 intubate")**; **9–12 moderate, 13–15 mild**; motor response best predictor.
- **Pupils** – normal **2–5 mm (PERRLA)**; **unilateral fixed dilated pupil – ipsilateral CN III compression (uncal herniation)**; **pin-point – pontine haemorrhage, opioids**.
- **Decorticate (flexor) posture** – lesion above midbrain; **decerebrate (extensor) posture** – brainstem/ midbrain lesion (worse prognosis).

**♫ Yaad karne ki trick**

- Meninges – **"DAP"** – Dura, Arachnoid, Pia (outer → inner).
- Spinal roots – **"DAVE"** – Dorsal Afferent, Ventral Efferent.
- Parasympathetic cranial nerves – **"1973"** – X, IX, VII, III.
- Diaphragm – **"C3, 4, 5 keeps the diaphragm alive"**.
- ANS sex – **"Point (Parasympathetic – erection) & Shoot (Sympathetic – ejaculation)"**.
- GCS – **E4 V5 M6**; **"8 – intubate"**.

### PYQ style

- Structural & functional unit of nervous system – **neuron**.
- Myelin in CNS – **oligodendrocytes**; PNS – **Schwann cells**.
- Macrophages of brain – **microglia**; BBB – **astrocytes**.
- Resting membrane potential – **−70 mV**.
- Main inhibitory neurotransmitter – **GABA**; excitatory – **glutamate**.
- Parkinson's – **↓ dopamine (substantia nigra)**; Alzheimer's – **↓ ACh**.
- Largest part of brain – **cerebrum**; balance – **cerebellum**.
- Vital centres – **medulla oblongata**.
- Temperature regulation, hunger, thirst – **hypothalamus**.
- Sensory relay – **thalamus**.
- Motor speech – **Broca's area (frontal)**; understanding speech – **Wernicke's (temporal)**.
- Vision – **occipital lobe**; hearing – **temporal lobe**.
- CSF formed by – **choroid plexus**; absorbed by – **arachnoid villi**; daily – **500 ml**; total – **150 ml**.
- Spinal cord ends at – **L1–L2 (adult)**, **L3 (newborn)**.
- LP site – **L3–L4 / L4–L5**.
- Knee jerk – **L3–L4 (monosynaptic)**.
- Cranial nerves – **12 pairs**; spinal – **31 pairs**.
- Largest cranial nerve – **trigeminal (V)**; longest – **vagus (X)**; smallest – **trochlear (IV)**.
- Bell's palsy – **facial (VII)**; ptosis & dilated pupil – **oculomotor (III)**.
- Lateral rectus – **abducens (VI)**; superior oblique – **trochlear (IV)**.
- Tongue movements – **hypoglossal (XII)**; taste anterior ⅔ – **facial (VII)**.
- Largest nerve – **sciatic**; wrist drop – **radial**; claw hand – **ulnar**; carpal tunnel – **median**; foot drop – **common peroneal**.
- Sympathetic outflow – **thoracolumbar (T1–L2)**; parasympathetic – **craniosacral**.
- Sympathetic effect on pupil – **dilation**.
- GCS range – **3–15**; coma – **≤ 8**.
- Cushing's triad – **hypertension, bradycardia, irregular breathing**.

# 10 Special Senses & Integumentary System

## 10.1 The Eye

- **Eyeball** – ~**24 mm** diameter; in **orbit** (7 bones); **protective structures** – eyelids, eyelashes, conjunctiva, **lacrimal apparatus (lacrimal gland – upper outer orbit → puncta → canaliculi → lacrimal sac → nasolacrimal duct → inferior meatus of nose)**; tears contain **lysozyme**.
- **Extraocular muscles (6)** – 4 recti (superior, inferior, medial, lateral) + 2 obliques; **"LR6 SO4, rest 3"** – lateral rectus by VI, superior oblique by IV, others by III.

| Layer | Parts | Points |
|---|---|---|
| **Outer fibrous** | **Sclera** (white, posterior 5/6), **cornea** (transparent, anterior 1/6) | **Cornea – avascular (nourished by aqueous & tears), most refractive surface (~⅔ of power), highly sensitive (CN V) – corneal transplant success**; limbus – junction |
| **Middle vascular (uvea)** | **Choroid** (vascular – nourishes retina), **ciliary body** (ciliary muscle – **accommodation**; ciliary processes – **aqueous humour**), **iris** (coloured; **pupil** – sphincter pupillae (parasympathetic – constriction), dilator pupillae (sympathetic – dilation)) | Uveitis |
| **Inner nervous** | **Retina** – **rods & cones** | – |

### Retina

- **Rods (~120 million)** – **dim light/ night (scotopic) vision**, peripheral, **rhodopsin (visual purple – needs vitamin A → night blindness in deficiency)**.
- **Cones (~6–7 million)** – **bright light, colour vision (photopic) & sharp vision**; 3 types (red, green, blue); **colour blindness – X-linked (red-green most common)** – **Ishihara chart**.
- **Macula lutea** – yellow spot; **fovea centralis – only cones – point of sharpest vision (maximum acuity)**.
- **Optic disc – "blind spot" – no photoreceptors** (optic nerve exit); **papilloedema – ↑ ICP**.
- **Light pathway** – cornea → aqueous humour → pupil → lens → vitreous humour → retina.

### Lens, humours & accommodation

- **Lens** – **biconvex, transparent, avascular, elastic**; held by **suspensory ligaments (zonules)**; **opacity → cataract (commonest cause of blindness in India)**.
- **Aqueous humour** – **formed by ciliary processes** → posterior chamber → pupil → anterior chamber → **drained via trabecular meshwork & canal of Schlemm** (at angle); maintains **IOP 10–21 mmHg (normal)**; ↑ IOP – **glaucoma** (tonometry).
- **Vitreous humour** – jelly behind lens; maintains shape; **not replaced**.
- **Accommodation (near vision)** – **ciliary muscle contracts → zonules relax → lens becomes more convex**, **pupil constricts, eyes converge** ("near triad"); **presbyopia** – age-related loss (~40+ yrs) – **convex lens**.

| Refractive error | Image | Correction |
|---|---|---|
| **Myopia (short-sight)** | Before retina (long eyeball) | **Concave (minus/ diverging) lens** |
| **Hypermetropia (long-sight)** | Behind retina (short eyeball) | **Convex (plus/ converging) lens** |
| **Astigmatism** | Irregular corneal curvature | **Cylindrical lens** |
| **Presbyopia** | Loss of accommodation (age) | **Convex lens (reading glasses)** |

- **Visual pathway** – retina → optic nerve → **optic chiasma (nasal fibres cross)** → optic tract → **lateral geniculate body (thalamus)** → optic radiation → **visual cortex (occipital)**; **pituitary tumour → bitemporal hemianopia**.
- **Normal visual acuity – 6/6 (Snellen chart at 6 m)**; **blindness (WHO/ India) – visual acuity < 3/60 in better eye**; near vision – Jaeger chart.
- **Light reflex** – light in one eye → **both pupils constrict (direct & consensual)**; afferent **II**, efferent **III**.
- **Vitamin A deficiency** – **night blindness (earliest) → conjunctival xerosis → Bitot's spots → corneal xerosis → keratomalacia**.

## 10.2 The Ear

| Part | Structures | Function |
|---|---|---|
| **External ear** | **Pinna (elastic cartilage)**, **external auditory canal (~2.5 cm, S-shaped; ceruminous glands – wax)** | Collects sound |
| **Middle ear (tympanic cavity)** | **Tympanic membrane (eardrum – pearly grey, cone of light antero-inferior)**, **ossicles – malleus (hammer), incus (anvil), stapes (stirrup – smallest bone; footplate on oval window)**; **Eustachian (auditory) tube – connects to nasopharynx – equalises pressure** (shorter, wider, horizontal in children → otitis media); muscles – **stapedius (smallest muscle – facial nerve)**, tensor tympani | **Amplifies (~20 ×) & transmits sound** |
| **Inner ear (labyrinth)** | **Cochlea (2¾ turns – organ of Corti – hearing receptors)**; **vestibule (utricle & saccule – linear acceleration & static balance)**; **semicircular canals (3 – rotational/ angular acceleration – dynamic balance)**; fluids – **perilymph (like ECF) & endolymph (high $K^+$)** | **Hearing & balance** |

- **Hearing pathway** – sound → pinna → canal → tympanic membrane → ossicles → **oval window** → cochlear fluid → **basilar membrane → hair cells of organ of Corti → cochlear nerve (VIII) → auditory cortex (temporal)**; **round window** – releases pressure.
- **Human audible range – 20–20,000 Hz**; **speech 500–2000 Hz**; normal conversation **~60 dB**; **noise > 85 dB prolonged → noise-induced hearing loss (4000 Hz notch)**; pain at ~120–130 dB.
- **Tuning fork tests (512 Hz)** – **Rinne** (normal AC > BC – positive), **Weber** (lateralises to **affected ear in conductive loss**, to **better ear in sensorineural loss**).
- **Conductive deafness** – external/ middle ear (wax, otitis media, otosclerosis); **sensorineural** – cochlea/ nerve (presbycusis, noise, ototoxic drugs, rubella).
- **Ear drops/ examination** – **adult pinna up & back; child < 3 yrs down & back**.
- **Newborn hearing screening** – OAE/ BERA.

## 10.3 Smell & Taste

- **Olfaction** – **olfactory epithelium in roof of nasal cavity** → olfactory receptor cells (**bipolar neurons – the only neurons regularly replaced**) → **olfactory bulb** → olfactory tract → **temporal lobe (uncus)/ limbic system**; **only sense not relayed via thalamus**; **adaptation fast**; anosmia (COVID, head injury).
- **Taste (gustation)** – **taste buds (~10,000)** on papillae (**circumvallate – largest, most taste buds**, fungiform, foliate; **filiform – no taste buds, most numerous**); **5 basic tastes – sweet, salty, sour, bitter, umami**; nerves – **VII (anterior ⅔ via chorda tympani), IX (posterior ⅓), X (epiglottis)**; taste closely linked with smell (flavour); taste cortex – **insula/ parietal**.

## 10.4 Skin (Integumentary System)

- **Largest organ of body** – **~1.5–2 m², ~15–16 % body weight**; thickest – **palms & soles**; thinnest – **eyelids**.

| Layer | Features |
|---|---|
| **Epidermis** (stratified squamous keratinised, **avascular**) | Layers (deep → surface) – **stratum basale (germinativum – dividing cells, melanocytes), spinosum, granulosum, lucidum (only in thick skin – palms/ soles), corneum (dead keratinised – protection)** – "**Come, Let's Get Sun Burned**" (surface → deep); cells – **keratinocytes (90 %), melanocytes (melanin – skin colour, UV protection), Langerhans cells (immune), Merkel cells (touch)**; renewed every ~4 weeks |
| **Dermis** (connective tissue – collagen & elastin) | **Papillary** (fingerprints, Meissner's corpuscles) & **reticular** layers; **blood vessels, nerves, hair follicles, sweat & sebaceous glands**, receptors |
| **Hypodermis (subcutaneous)** | **Adipose & loose connective tissue** – insulation, energy store; **SC injection site** |

- **Glands** – **eccrine sweat glands** (all over, **most on palms, soles, forehead** – thermoregulation; watery), **apocrine sweat glands** (axilla, groin, areola – active after puberty; **body odour** by bacteria), **sebaceous glands (sebum – holocrine; absent on palms & soles; acne)**, **ceruminous (ear wax)**, **mammary (modified sweat glands)**.
- **Functions of skin** – **protection** (barrier, UV, microbes – acid mantle pH 4.5–5.5), **thermoregulation**, **sensation** (touch – Meissner's; pressure/ vibration – **Pacinian**; pain – free nerve endings; cold – Krause; heat – Ruffini), **excretion** (sweat – water, salt, urea), **vitamin D synthesis (7-dehydrocholesterol + UVB → cholecalciferol)**, absorption (lipid-soluble drugs – transdermal patches), **fluid balance**, immunity, storage (fat).
- **Skin colour** – **melanin** (brown-black; **same number of melanocytes in all races – activity differs**), **carotene** (yellow), **haemoglobin** (pink); **albinism – absence of melanin (tyrosinase deficiency)**; **vitiligo** – loss of melanocytes.
- **Hair** – **keratin**; shaft, root, follicle, **arrector pili (goosebumps – sympathetic)**; growth phases anagen (growth), catagen, telogen; **lanugo (fine fetal hair)**, vernix caseosa (cheesy coating of newborn).
- **Nails** – keratin plates; **lunula (white crescent)**; **grow ~3 mm/month (fingernails faster than toenails)**; **clubbing** (chronic hypoxia, lung cancer, bronchiectasis, cyanotic heart disease), **koilonychia (spoon nails – iron deficiency)**, **capillary refill < 2 (–3) s**.

## Body temperature regulation

- **Centre – hypothalamus (anterior – heat loss; posterior – heat conservation/ production)**.
- **Normal core temperature ~37 °C (98.6 °F)**; range **36.5–37.5 °C**; **oral 37 °C, rectal ~0.5 °C higher (most accurate of common sites – core), axillary ~0.5 °C lower**; **lowest ~early morning (2–6 AM), highest ~late afternoon/ evening (circadian)**; ↑ after ovulation (**progesterone, ~0.3–0.5 °C**).
- **Heat production** – **metabolism (liver most at rest)**, **muscle activity/ shivering (most during exercise)**, **brown fat (non-shivering thermogenesis – newborns; between scapulae, around kidneys)**, thyroxine, adrenaline.
- **Heat loss** – **radiation (~60 % – maximum at rest)**, **conduction**, **convection**, **evaporation (sweating – only way when environment hotter than body; main in exercise)**.
- **Fever (pyrexia)** – > 37.5–38 °C (pyrogens – IL-1, PGE2 reset hypothalamic set point); **hyperpyrexia > 41–41.5 °C (106 °F)**; **hypothermia < 35 °C**; newborns at risk of hypothermia (large surface area, little SC fat) – **"warm chain"**.
- **°F = (°C × 9/5) + 32**; **°C = (°F − 32) × 5/9**.

**♫ Yaad karne ki trick**

- Eye muscles – **"LR6 SO4, rest 3"**.
- Myopia – **"Minus for Myopia"** (concave).
- Epidermis surface → deep – **"Come, Let's Get Sun Burned"** – corneum, lucidum, granulosum, spinosum, basale.
- Ossicles – **"MIS"** – Malleus, Incus, Stapes (outside → in).
- Weber in conductive loss – **"Weber goes to the bad ear in conductive"**.

### PYQ style

- Transparent avascular anterior part – **cornea**; white of eye – **sclera**.
- Most refractive structure – **cornea**.
- Pigment of rods – **rhodopsin (vitamin A)**; night vision – **rods**; colour vision – **cones**.
- Sharpest vision – **fovea centralis**; blind spot – **optic disc**.
- Aqueous humour formed by – **ciliary processes**; drained by – **canal of Schlemm**.
- Normal IOP – **10–21 mmHg**; ↑ IOP – **glaucoma**.
- Accommodation – **ciliary muscle**.
- Myopia corrected by – **concave lens**; hypermetropia & presbyopia – **convex**; astigmatism – **cylindrical**.
- Normal visual acuity – **6/6**; test – **Snellen chart**; colour blindness – **Ishihara chart**.
- Commonest cause of blindness in India – **cataract**.
- Earliest sign of vitamin A deficiency – **night blindness**.
- Smallest bone – **stapes**; smallest muscle – **stapedius**.
- Eustachian tube connects – **middle ear to nasopharynx**.
- Organ of hearing – **organ of Corti (cochlea)**; balance – **semicircular canals & vestibule**.
- Audible range – **20–20,000 Hz**.
- Weber lateralises to affected ear – **conductive deafness**.
- Only sense not relayed via thalamus – **smell**.
- Largest organ – **skin**; thickest skin – **palms & soles**.
- Skin pigment – **melanin**; layer with dividing cells – **stratum basale**.
- Stratum lucidum present in – **palms & soles**.
- Vitamin synthesised in skin – **vitamin D**.
- Sebaceous glands absent on – **palms & soles**.
- Spoon-shaped nails – **iron deficiency**; clubbing – **chronic hypoxia**.
- Temperature-regulating centre – **hypothalamus**.
- Maximum heat loss at rest – **radiation**; in hot environment – **evaporation**.
- Non-shivering thermogenesis in newborn – **brown fat**.
- Lowest body temperature – **early morning**.

# 11 Endocrine System

## 11.1 Hormones – Basics

- **Endocrine glands – ductless**, secrete **hormones** directly into blood; **"Father of Endocrinology" – Thomas Addison**; **first hormone discovered – secretin (1902)**; **"hormone" term – Starling (1905)**.
- **Chemical types** – **peptide/ protein** (insulin, GH, ADH, oxytocin, PTH – water-soluble, **cell-membrane receptors, 2nd messenger cAMP**), **steroids** (cortisol, aldosterone, sex hormones, vitamin D – lipid-soluble, **intracellular/ nuclear receptors**), **amines** (adrenaline, noradrenaline, **thyroid hormones – act like steroids (nuclear receptors)**, melatonin).
- **Regulation** – mostly **negative feedback** (e.g. ↑ T4 → ↓ TSH); positive – oxytocin in labour, LH surge.
- **Master gland – pituitary**; **"master of master" – hypothalamus**; **largest endocrine gland – thyroid**; **mixed glands – pancreas, gonads**.

## 11.2 Hypothalamus & Pituitary

- **Hypothalamus** – **releasing & inhibiting hormones** via **hypothalamo-hypophyseal portal system** to anterior pituitary – **TRH, CRH, GnRH, GHRH, somatostatin (GH-inhibiting), dopamine (prolactin-inhibiting)**; **makes ADH & oxytocin** (supraoptic & paraventricular nuclei) – transported along axons to **posterior pituitary (stored & released)**.
- **Pituitary (hypophysis)** – **pea-sized (~0.5 g), in sella turcica of sphenoid**, attached by infundibulum; **anterior (adenohypophysis – glandular)** & **posterior (neurohypophysis – neural)**.

| Hormone (anterior pituitary) | Action | Hypo | Hyper |
|---|---|---|---|
| **Growth hormone (GH/ somatotropin)** | Growth of bones & tissues (via **IGF-1 from liver**), protein synthesis, ↑ blood glucose (anti-insulin); **secreted mostly during deep sleep** | **Dwarfism (proportionate)** in children | **Gigantism (before epiphyseal fusion)**, **acromegaly (adults – large hands, feet, jaw)** |
| **TSH (thyrotropin)** | Thyroid hormone release | Hypothyroidism | Hyperthyroidism |
| **ACTH (corticotropin)** | Cortisol release (adrenal cortex) | Secondary adrenal insufficiency | **Cushing's disease** |
| **FSH** | **Female – follicle growth, oestrogen; male – spermatogenesis (Sertoli cells)** | Infertility | – |
| **LH (ICSH in males)** | **Female – ovulation (LH surge), corpus luteum; male – testosterone (Leydig cells)** | Infertility | – |
| **Prolactin** | **Milk production (lactogenesis)** | Lactation failure (**Sheehan's syndrome – postpartum pituitary necrosis after PPH**) | **Galactorrhoea, amenorrhoea, infertility** |
| MSH | Melanin (pigmentation) | – | – |

| Hormone (posterior pituitary) | Action | Disorder |
|---|---|---|
| **ADH (vasopressin)** | **Water reabsorption in collecting ducts (aquaporins)**, vasoconstriction | **Deficiency → diabetes insipidus (polyuria, dilute urine, low sp. gravity < 1.005, thirst)**; **excess → SIADH (hyponatraemia, concentrated urine, water retention)** |
| **Oxytocin ("love/ cuddle hormone")** | **Uterine contraction (labour – Ferguson reflex, positive feedback)**, **milk ejection (let-down reflex – suckling)**, bonding | Used in induction & PPH |

- **Pituitary tumour** – **bitemporal hemianopia** (optic chiasma pressure).

## 11.3 Thyroid Gland

- **Largest endocrine gland (~20–25 g)**; **butterfly-shaped**, **2 lobes + isthmus (over 2nd–4th tracheal rings)**, front of neck **C5–T1**; **moves on swallowing** (pretracheal fascia); highly vascular; **functional unit – follicle (colloid – thyroglobulin)**.
- **Follicular cells** – **T4 (thyroxine – ~90 % secreted, prohormone)** & **T3 (triiodothyronine – ~3–5 × more potent, active form; most T4 → T3 in peripheral tissues)**; need **iodine (daily requirement ~150 µg adult, 250 µg pregnancy/ lactation)**; **mostly bound to TBG** in blood.
- **Parafollicular (C) cells** – **calcitonin (↓ blood calcium – inhibits osteoclasts)**; tumour marker for medullary carcinoma.
- **Actions of T3/T4** – **↑ BMR & $O_2$ consumption, heat production (calorigenic)**, ↑ heart rate & CO, **growth & brain development (essential in fetus & infancy)**, ↑ metabolism of carbs, fats, proteins, ↑ sensitivity to catecholamines.

| Condition | Features |
|---|---|
| **Hypothyroidism – infant: congenital hypothyroidism (cretinism)** | **Mental retardation, short stature, large tongue (macroglossia), umbilical hernia, prolonged jaundice, constipation, coarse cry → newborn screening (TSH, heel prick) & early thyroxine** |
| **Hypothyroidism – adult: myxoedema** | **Weight gain, cold intolerance, bradycardia, lethargy, dry skin, hair loss, constipation, menorrhagia, slow reflexes, non-pitting oedema**; **Hashimoto's thyroiditis – commonest cause (autoimmune)** in iodine-sufficient areas; worldwide – iodine deficiency |
| **Hyperthyroidism (thyrotoxicosis)** | **Weight loss despite appetite, heat intolerance, tachycardia/ AF, tremors, sweating, anxiety, diarrhoea, amenorrhoea**; **Graves' disease (commonest – TSH-receptor stimulating antibodies; exophthalmos, pretibial myxoedema, diffuse goitre)** |
| **Goitre** | Enlarged thyroid – **simple/ endemic (iodine deficiency – sub-Himalayan "goitre belt")**; **NIDDCP – iodised salt (≥ 15 ppm at consumer, 30 ppm at production)** |
| **Thyroid storm** | Life-threatening hyperthyroidism – hyperpyrexia, tachycardia |

- **Normal values** – **TSH ~0.4–4.0 mIU/L**, free T4 ~0.8–1.8 ng/dl, T3 ~80–200 ng/dl; **TSH – best screening test**.

## 11.4 Parathyroid Glands

- **4 small glands (2 superior, 2 inferior)** on **posterior surface of thyroid**; **chief cells – parathormone (PTH)**.
- **PTH ↑ blood calcium** – **bone resorption (osteoclasts)**, **↑ renal $Ca^{2+}$ reabsorption (DCT)**, **↓ phosphate reabsorption (phosphaturia)**, **activates vitamin D (1α-hydroxylase in kidney) → ↑ intestinal $Ca^{2+}$ absorption**; secretion controlled by **serum $Ca^{2+}$ (calcium-sensing receptor)**.
- **Hypoparathyroidism** (commonest cause – **accidental removal/ injury during thyroidectomy**) → **hypocalcaemia → tetany, Chvostek's sign (facial twitch on tapping facial nerve), Trousseau's sign (carpal spasm with BP cuff), laryngospasm, seizures, prolonged QT → IV calcium gluconate (10 %) at bedside after thyroidectomy**.
- **Hyperparathyroidism → hypercalcaemia – "stones (renal), bones (pain, fractures), groans (abdominal – peptic ulcer, pancreatitis, constipation), moans (psychiatric)"**, polyuria.

## 11.5 Adrenal (Suprarenal) Glands

- **Pair, on upper poles of kidneys**, retroperitoneal; **right – pyramidal, left – semilunar**; ~4–5 g each.
- **Adrenal cortex (outer, ~80–90 %, from mesoderm, steroid hormones – essential for life)** – 3 zones "**GFR – Salt, Sugar, Sex – the deeper you go the sweeter it gets**":

| Zone | Hormone | Action |
|---|---|---|
| **Zona glomerulosa (outer)** | **Mineralocorticoids – aldosterone** | **$Na^+$ & water retention, $K^+$ & $H^+$ excretion (DCT)** → ↑ BP; regulated by **RAAS (angiotensin II) & ↑ $K^+$** (not much by ACTH); excess → **Conn's syndrome (hypertension + hypokalaemia)** |
| **Zona fasciculata (middle, largest)** | **Glucocorticoids – cortisol (hydrocortisone)** | **"Stress hormone" – ↑ blood glucose (gluconeogenesis), protein catabolism, fat redistribution, anti-inflammatory & immunosuppressive**, maintains BP (permissive for catecholamines); **diurnal rhythm – highest ~8 AM (early morning), lowest at midnight**; regulated by **ACTH** |
| **Zona reticularis (inner)** | **Androgens (DHEA)** | Pubic/ axillary hair (females), libido |

- **Cushing's syndrome (cortisol excess) – moon face, buffalo hump, truncal obesity with thin limbs, purple striae, hypertension, hyperglycaemia, osteoporosis, hirsutism, thin skin, poor wound healing, infections, hypokalaemia**; commonest cause – **iatrogenic (long-term steroids)**; Cushing's disease = pituitary ACTH-secreting adenoma.
- **Addison's disease (primary adrenal insufficiency – ↓ cortisol & aldosterone) – weakness, weight loss, hypotension, hypoglycaemia, hyponatraemia, hyperkalaemia, hyperpigmentation (↑ ACTH/ MSH – skin creases, buccal mucosa)**, salt craving; **commonest cause – autoimmune (TB in India historically)**; **Addisonian crisis – IV hydrocortisone + saline**.
- **Congenital adrenal hyperplasia (21-hydroxylase deficiency)** – ambiguous genitalia in female newborns, salt-wasting.
- **Adrenal medulla (inner, from neural crest; modified sympathetic ganglion) – chromaffin cells – adrenaline (epinephrine ~80 %) & noradrenaline (~20 %) – "fight or flight"**: ↑ HR, BP, glucose, bronchodilation, pupil dilation; **not essential for life**; tumour – **phaeochromocytoma (paroxysmal hypertension, headache, sweating, palpitations; urinary VMA/ metanephrines)**.

## 11.6 Pancreas (Islets of Langerhans)

| Cell | Hormone | Action |
|---|---|---|
| **α (alpha) ~20 %** | **Glucagon** | **↑ blood glucose – glycogenolysis, gluconeogenesis** (liver); released in hypoglycaemia, fasting |
| **β (beta) ~70 %** | **Insulin** (51 amino acids – 2 chains A & B; from proinsulin → **C-peptide**) | **↓ blood glucose – ↑ glucose uptake into muscle & fat (GLUT-4), glycogenesis, lipogenesis, protein synthesis; inhibits gluconeogenesis**; **only hypoglycaemic hormone**; **drives $K^+$ into cells** |
| **δ (delta)** | **Somatostatin** | Inhibits insulin & glucagon, GH |
| **PP (F) cells** | Pancreatic polypeptide | – |

- **Hyperglycaemic hormones** – **glucagon, cortisol, GH, adrenaline, thyroxine** ("counter-regulatory").
- **Diabetes mellitus** – **type 1 (autoimmune β-cell destruction – absolute insulin deficiency; young; DKA)**; **type 2 (insulin resistance + relative deficiency; adults, obesity; most common ~90–95 %)**; **gestational**; **diagnosis – FBS ≥ 126 mg/dl, 2-hr PPBS/ OGTT (75 g) ≥ 200, random ≥ 200 with symptoms, HbA1c ≥ 6.5 %**; classic triad – **polyuria, polydipsia, polyphagia**.
- **Insulin discovered – Banting & Best (1921)**; **World Diabetes Day – 14 November (Banting's birthday)**.

## 11.7 Other Endocrine Tissues

| Gland/ tissue | Hormone | Key point |
|---|---|---|
| **Pineal (epiphysis cerebri)** | **Melatonin** | **Circadian rhythm, sleep; secreted in darkness**; "third eye"; calcifies with age |
| **Thymus** | **Thymosin, thymopoietin** | **T-cell maturation**; **largest at puberty, then involutes (fat)**; in anterior mediastinum |
| **Ovaries** | **Oestrogen (Graafian follicle), progesterone (corpus luteum)**, inhibin, relaxin | Female sex characters, menstrual cycle, pregnancy maintenance |
| **Testes** | **Testosterone (Leydig/ interstitial cells – LH)**; inhibin (Sertoli) | Male characters, spermatogenesis |
| **Placenta** | **hCG (basis of pregnancy test – detectable ~8–10 days after conception; peaks ~8–10 wks; maintains corpus luteum)**, **hPL (human placental lactogen – anti-insulin)**, oestrogen (oestriol), progesterone, relaxin | Temporary endocrine organ |
| **Kidney** | **Erythropoietin, renin, calcitriol** | – |
| **Heart (atria)** | **ANP (atrial natriuretic peptide)** | ↓ BP, ↑ $Na^+$ excretion |
| **GI tract** | **Gastrin, secretin, CCK, GIP, ghrelin (hunger)** | – |
| **Adipose tissue** | **Leptin (satiety)** | – |

| Gland/ tissue | Hormone | Key point |
|---|---|---|
| **Liver** | **IGF-1 (somatomedin), angiotensinogen, thrombopoietin** | – |
| **Local hormones** | **Prostaglandins** (inflammation, pain, fever, labour, gastric protection), **histamine, bradykinin, serotonin** | Act locally (autocrine/ paracrine) |

### ♫ Yaad karne ki trick

- Anterior pituitary hormones – **"FLAT PiG"** – FSH, LH, ACTH, TSH, Prolactin, GH.
- Adrenal cortex – **"GFR = Salt, Sugar, Sex"** – glomerulosa (aldosterone), fasciculata (cortisol), reticularis (androgens).
- Adrenal layers outer → inner – **"GFR-M"** + medulla (adrenaline).
- Hypercalcaemia – **"Stones, bones, groans, moans"**.
- Posterior pituitary – **"ADH & Oxytocin – made in hypothalamus, stored in posterior"**.

### ✎ PYQ style

- Master gland – **pituitary**; controls pituitary – **hypothalamus**.
- Largest endocrine gland – **thyroid**.
- Pituitary lies in – **sella turcica (sphenoid)**.
- GH excess in adult – **acromegaly**; in child – **gigantism**; deficiency – **dwarfism**.
- ADH deficiency – **diabetes insipidus**; excess – **SIADH**.
- ADH & oxytocin synthesised in – **hypothalamus**; released from – **posterior pituitary**.
- Milk ejection – **oxytocin**; milk production – **prolactin**.
- Sheehan's syndrome – **postpartum pituitary necrosis**.
- Calcitonin – **thyroid C (parafollicular) cells – ↓ calcium**.
- Active thyroid hormone – **T3**; most secreted – **T4**.
- Congenital hypothyroidism – **cretinism (mental retardation)**; adult – **myxoedema**.
- Goitre – **iodine deficiency**; exophthalmos – **Graves' disease**.
- PTH – **↑ calcium**; deficiency after thyroidectomy – **tetany**.
- Chvostek & Trousseau signs – **hypocalcaemia**.
- Aldosterone – **zona glomerulosa**; cortisol – **zona fasciculata**.
- Moon face, buffalo hump – **Cushing's syndrome**.
- Hyperpigmentation, hypotension, hyperkalaemia – **Addison's disease**.
- Adrenaline – **adrenal medulla**; phaeochromocytoma – **episodic hypertension**.
- Cortisol highest – **early morning**.
- Insulin – **β cells**; glucagon – **α cells**; only hypoglycaemic hormone – **insulin**.
- Melatonin – **pineal gland**; T-cell maturation – **thymus**.
- Pregnancy test hormone – **hCG**.
- Testosterone – **Leydig cells**.
- Erythropoietin – **kidney**; ANP – **heart (atria)**.
- World Diabetes Day – **14 November**.

# 12 Reproductive System

## 12.1 Male Reproductive System

- **Testes** – **in scrotum (temperature ~2–3 °C below body – needed for spermatogenesis)**; ~4–5 cm, oval; covered by **tunica vaginalis & tunica albuginea**; **descend through inguinal canal by 7th–8th month / at birth (gubernaculum)**; failure → **cryptorchidism (undescended – infertility, malignancy risk; orchiopexy by ~1 yr)**.
- **Seminiferous tubules** – **spermatogenesis**; **Sertoli cells (nurse/ sustentacular – support & nourish sperm, blood-testis barrier, inhibin, ABP – FSH-driven)**; **Leydig (interstitial) cells – testosterone (LH)**.
- **Spermatogenesis** – **~64–74 days**; spermatogonium (46) → primary spermatocyte → (meiosis I) → secondary spermatocyte (23) → (meiosis II) → spermatid → **spermiogenesis** → **spermatozoon**; **1 primary spermatocyte → 4 sperms**; begins at puberty, lifelong.
- **Sperm** – **head (nucleus + acrosome – enzymes – hyaluronidase to penetrate ovum)**, **middle piece (mitochondria – energy)**, tail (flagellum – motility); length ~55–60 µm; **viable in female tract ~48–72 hrs (up to 5 days)**.
- **Duct system** – **seminiferous tubules → rete testis → efferent ductules → epididymis (~6 m coiled; sperm maturation & storage, gain motility) → vas (ductus) deferens (~45 cm – passes through inguinal canal; cut in vasectomy) → joins seminal vesicle duct = ejaculatory duct → prostatic urethra**.
- **Accessory glands** – **seminal vesicles (~60–70 % of semen; alkaline, fructose – energy for sperm, prostaglandins)**, **prostate (~20–30 %; slightly acidic/ milky, citric acid, PSA, acid phosphatase; surrounds urethra below bladder – BPH (elderly) → urinary obstruction)**, **bulbourethral (Cowper's) glands (pre-ejaculate mucus – lubrication)**.
- **Semen (WHO 2021)** – **volume ≥ 1.4 ml (~2–5 ml)**, **sperm concentration ≥ 16 million/ml (~39 million total)**, **total motility ≥ 42 %**, progressive ≥ 30 %, normal forms ≥ 4 %; pH 7.2–8.0; **azoospermia** (none), **oligospermia** (low count).
- **Penis** – **corpora cavernosa (2) & corpus spongiosum (1 – urethra; glans)**; **erection – parasympathetic (NO, vasodilation)**, **ejaculation – sympathetic**; **prepuce (foreskin)** – phimosis.
- **Testosterone** – **male secondary sex characters**, muscle & bone mass, libido, spermatogenesis (with FSH), voice change (larynx), **epiphyseal closure**, ↑ RBC.

## 12.2 Female Reproductive System – Organs

### External genitalia (vulva)

- **Mons pubis**, **labia majora** (homologous to scrotum), **labia minora**, **clitoris (erectile – homologous to penis)**, **vestibule** (urethral & vaginal openings), **Bartholin's glands (greater vestibular – lubrication; cyst/ abscess)**, Skene's glands, hymen, **perineum (perineal body – episiotomy, tears)**.

### Internal organs

| Organ | Features |
|---|---|
| **Vagina** | **~7–10 cm fibromuscular canal** (anterior wall ~7.5 cm, posterior ~9 cm); **acidic pH 3.8–4.5 (Döderlein's bacilli/ lactobacilli – lactic acid from glycogen; oestrogen-dependent)**; **posterior fornix deepest** (culdocentesis); stratified squamous non-keratinised; no glands |
| **Uterus** | **Pear-shaped, hollow, muscular**; **~7.5 × 5 × 2.5 cm (3 × 2 × 1 inch), ~50–60 g (non-pregnant) → ~1000 g at term**; capacity ~5–10 ml → ~5 L at term; **normal position – anteverted & anteflexed**; parts – **fundus, body (corpus), isthmus (→ lower uterine segment in pregnancy), cervix**; layers – **perimetrium, myometrium (smooth muscle – "living ligatures" – oblique fibres in middle layer control bleeding after delivery), endometrium (functional layer – shed in menses; basal layer – regenerates)** |
| **Cervix** | **~2.5 cm**; **internal os & external os** (round in nulliparous, transverse slit in parous); **transformation zone (squamocolumnar junction) – site of cervical cancer → Pap smear**; cervical mucus changes (spinnbarkeit/ ferning at ovulation) |
| **Fallopian (uterine) tubes** | **~10–12 cm**; parts (lateral → medial) – **fimbriae, infundibulum, ampulla (widest, longest – site of fertilisation & commonest site of ectopic pregnancy), isthmus (narrowest – tubectomy site), interstitial (intramural)**; **ciliated columnar epithelium** + peristalsis move ovum |
| **Ovaries** | **Almond-shaped, ~3 × 2 × 1 cm**; on **ovarian fossa**; ligaments – ovarian & suspensory (infundibulopelvic – carries ovarian vessels); **cortex (follicles) & medulla**; produce **ova, oestrogen, progesterone**, inhibin, androgens |

- **Uterine supports** – **pelvic floor (levator ani)**, **cardinal (Mackenrodt's/ transverse cervical) ligaments – main support**, uterosacral ligaments, pubocervical; **round ligament (keeps anteversion; passes through inguinal canal)**; **broad ligament (peritoneal fold – not a true support)**; weakness → **uterine prolapse**.
- **Blood supply** – **uterine artery (from internal iliac) – crosses above ureter ("water under the bridge" – injury risk in hysterectomy)**; ovarian artery (from aorta).
- **Pelvis (female/ gynaecoid)** – **round inlet, wide subpubic arch (> 90°), shallow, ischial spines not prominent**; **true pelvis – inlet, cavity, outlet**; **diagonal conjugate ~12.5 cm (true/ obstetric conjugate ≈ diagonal − 1.5 cm ≈ 11 cm)**; **pouch of Douglas (rectouterine pouch) – lowest point of peritoneal cavity (fluid/ pus collects)**.

## 12.3 Oogenesis & Ovarian Cycle

- **Oogenesis begins before birth** – oogonia multiply in fetus; **primary oocytes arrested in prophase I** – **~7 million at 20 weeks fetal life (maximum) → ~1–2 million at birth → ~3–4 lakh at puberty → ~400–500 ovulated in lifetime**.
- **Meiosis I completes just before ovulation** (→ secondary oocyte + 1st polar body); **secondary oocyte arrested in metaphase II – completes meiosis II only if fertilised** (→ ovum + 2nd polar body); **1 primary oocyte → 1 ovum + 3 (or 2) polar bodies**.
- **Follicle development** – primordial → primary → secondary → **Graafian (mature, vesicular) follicle (~18–25 mm) → ovulation → corpus luteum (progesterone; lasts ~14 days if no pregnancy → corpus albicans)**; in pregnancy maintained by **hCG** till placenta takes over (~10–12 wks).
- **Ovum** – **largest human cell (~120–150 µm)**; **viable ~12–24 hrs after ovulation**; zona pellucida, corona radiata.

## 12.4 Menstrual Cycle

- **Menarche** – first menses (**~11–14 yrs in India**; average ~12–13); **menopause** – cessation for **12 consecutive months (~45–55 yrs; average ~51 worldwide, ~46–48 in India)**.
- **Normal cycle 28 days (21–35)**; **flow 3–5 days (2–7)**; **blood loss ~35 ml (20–80; > 80 ml – menorrhagia)**; **menstrual blood doesn't clot (fibrinolysins/ plasmin)**.

| Phase | Days (28-day cycle) | Ovary | Endometrium | Hormones |
|---|---|---|---|---|
| **Menstrual** | **1–5** | Corpus luteum regresses; new follicles recruited | **Functional layer shed** | **↓ oestrogen & progesterone** (trigger) |
| **Follicular/ Proliferative** | **6–14** | Follicle grows (FSH) → Graafian | **Proliferation – regeneration & thickening** | **Oestrogen ↑ (dominant)** |
| **Ovulation** | **~Day 14** (**14 days before next period**) | Graafian follicle ruptures | – | **LH surge (peak ~24–36 hrs before ovulation) – positive feedback by high oestrogen** |
| **Luteal/ Secretory** | **15–28** (constant **14 days**) | **Corpus luteum** | **Secretory – glands tortuous, glycogen-rich, oedematous – ready for implantation** | **Progesterone ↑ (dominant)**, oestrogen (2nd peak) |

- **Signs of ovulation** – **rise in BBT by ~0.3–0.5 °C (progesterone – thermogenic)**, **clear stretchy cervical mucus (spinnbarkeit; fern pattern)**, **mittelschmerz (mid-cycle pain)**, LH kits.
- **Fertile period** – ~**day 10–17** (5 days before to 1 day after ovulation); calendar/ standard days method (**days 8–19** – CycleBeads).
- **Oestrogen** – **proliferative phase**, female secondary characters (breast duct growth, fat distribution), **↑ cervical mucus (thin)**, maintains vaginal epithelium, bone protection, ↑ HDL, epiphyseal closure.
- **Progesterone ("pro-gestation" – pregnancy hormone)** – **secretory endometrium, maintains pregnancy (relaxes uterus – ↓ contractility)**, **thick cervical mucus**, **↑ BBT**, breast alveolar development, relaxes smooth muscle (constipation, heartburn, ureteric dilation in pregnancy).
- **Puberty sequence (girls)** – **thelarche (breast budding ~8–13 yrs – first sign) → pubarche → growth spurt → menarche**; Tanner staging.

## 12.5 Breast & Lactation

- **Modified sweat (apocrine) gland**; **over 2nd–6th ribs**, **axillary tail of Spence (upper outer quadrant – most glandular tissue – commonest site of breast cancer)**; **15–20 lobes** → lobules → **alveoli (milk secretion)** → lactiferous ducts → **lactiferous sinuses (under areola)** → nipple; **Montgomery's tubercles** (sebaceous glands on areola – enlarge in pregnancy); **Cooper's ligaments (suspensory – dimpling in cancer)**; **lymph drainage – ~75 % to axillary nodes**.
- **Lactation hormones** – **development: oestrogen (ducts), progesterone (alveoli), hPL, prolactin**; **milk production – prolactin** (inhibited during pregnancy by high oestrogen & progesterone; starts after placenta delivery); **milk ejection (let-down) – oxytocin**; **suckling stimulates both (neuroendocrine reflex)**; prolactin also suppresses ovulation (**LAM – lactational amenorrhoea method, 6 months, exclusive breastfeeding**).

- **Colostrum** – first **2–4 days**, **yellow, thick**, **rich in protein, IgA, vitamin A, WBCs**, less fat & sugar – **"first immunisation"**; transitional milk (~day 5–14) → mature milk; **foremilk (watery – thirst) & hindmilk (fat-rich – satiety, weight gain)**; daily output ~500–800 ml.

## 12.6 Fertilisation, Implantation & Placenta

- **Fertilisation** – fusion of sperm & ovum in **ampulla** of fallopian tube within ~**12–24 hrs of ovulation**; **capacitation** of sperm (in female tract) → **acrosome reaction** → zona pellucida penetration → **zona reaction (prevents polyspermy)** → **zygote (46 chromosomes; sex determined by sperm – X or Y)**.
- **Cleavage** – zygote → 2, 4, 8 cells → **morula (16 cells, ~day 3–4, enters uterus)** → **blastocyst (~day 5; inner cell mass → embryo; trophoblast → placenta)**.
- **Implantation** – **~day 6–7 after fertilisation (complete by ~day 10–12)**; usually **upper posterior wall of uterine body (fundus)**; endometrium now called **decidua** (basalis – maternal placenta, capsularis, parietalis/ vera); implantation bleeding (Hartman's sign).
- **Embryonic period – up to 8 weeks (organogenesis, 3–8 wks – most teratogen-sensitive)**; **fetal period – 9 weeks to birth**; **heart starts beating ~day 22 (5–6 weeks LMP on USG)**.
- **Germ layers** – **ectoderm (skin/ epidermis, nervous system, eye lens, enamel, hair, nails)**, **mesoderm (muscles, bones, heart, blood, kidneys, gonads, dermis)**, **endoderm (lining of GI & respiratory tracts, liver, pancreas, thyroid, bladder lining)**.
- **Placenta** – **discoid, ~15–20 cm diameter, ~2.5–3 cm thick, ~500 g (≈ ⅙ of baby's weight) at term**; **fetal surface – smooth, shiny (amnion), cord inserted (Schultze)**; **maternal surface – rough, 15–20 cotyledons (Duncan)**; functions – **respiratory, nutritive, excretory, barrier (partial), endocrine (hCG, hPL, oestrogen, progesterone), immunological (IgG transfer)**; formed fully by ~12 weeks.
- **Umbilical cord** – **~50 cm (30–100)**, **2 arteries + 1 vein**, **Wharton's jelly**.
- **Fetal membranes** – **amnion (inner)** & **chorion (outer)**; **amniotic fluid ~800–1000 ml at 36–38 wks** (polyhydramnios > 2000 ml, oligohydramnios < 200–300 ml / AFI < 5 cm); **clear, slightly alkaline (pH ~7–7.5 – nitrazine test turns blue)**; functions – cushion, temperature, free movement, lung development, prevents adhesions; **fetal urine is major source in late pregnancy**.
- **Duration of pregnancy – 280 days (40 weeks) from LMP**; **266 days (38 weeks) from fertilisation**; **Naegele's rule – EDD = LMP + 9 months + 7 days**.

## 12.7 Menopause

- **Permanent cessation of menses (12 months amenorrhoea)** due to **ovarian follicle depletion**; **↓ oestrogen → ↑ FSH (diagnostic) & LH**.
- **Features** – **hot flushes (most common)**, night sweats, mood changes, insomnia, **vaginal dryness/ atrophy (pH rises)**, urinary symptoms, **osteoporosis (↑ fracture risk)**, **↑ cardiovascular risk (↓ HDL protection)**, skin changes.
- **Premature menopause** – before 40 yrs; **post-menopausal bleeding – always investigate (endometrial cancer)**.
- **Andropause** – gradual ↓ testosterone in men.

### Yaad karne ki trick

- Sperm pathway – **"SEVEn UP"** – **S**eminiferous tubules, **E**pididymis, **V**as deferens, **E**jaculatory duct, (**n**othing), **U**rethra, **P**enis.
- Fallopian tube parts (lateral → medial) – **"FIAII"** – Fimbriae, Infundibulum, Ampulla, Isthmus, Interstitial.
- Germ layers – **Ecto = Exterior (skin, nerves)**; **Endo = inner linings (gut)**; **Meso = Middle (muscle, bone, blood)**.
- Luteal phase – **"Luteal is Loyal – always 14 days"**.

### PYQ style

- Sperm formation – **seminiferous tubules**; testosterone – **Leydig cells**; nurse cells – **Sertoli**.
- Sperm maturation & storage – **epididymis**; cut in vasectomy – **vas deferens**.
- Largest contribution to semen – **seminal vesicles**; fructose source – **seminal vesicles**.
- Spermatogenesis needs temperature – **2–3 °C below body temperature**.
- Spermatogenesis duration – **~64–74 days**.
- Site of fertilisation – **ampulla of fallopian tube**; commonest ectopic – **ampulla**.
- Tubectomy site – **isthmus**.
- Normal uterus position – **anteverted & anteflexed**.
- Main uterine support – **cardinal (Mackenrodt's) ligament**.
- Non-pregnant uterus weight – **50–60 g**; at term – **~1 kg**.
- Vaginal pH – **3.8–4.5 (Döderlein's bacilli)**.
- Cervical cancer screening – **Pap smear (transformation zone)**.
- Ovulation occurs – **14 days before next menses**; due to – **LH surge**.
- Proliferative phase hormone – **oestrogen**; secretory phase – **progesterone**.
- BBT rise after ovulation – **progesterone**.
- Normal menstrual blood loss – **~35 ml (20–80)**.
- First sign of puberty in girls – **thelarche (breast budding)**.
- Milk production – **prolactin**; ejection – **oxytocin**.
- Colostrum rich in – **IgA, protein, vitamin A**.
- Commonest site of breast cancer – **upper outer quadrant**.
- Implantation – **day 6–7 after fertilisation**.
- Organogenesis period – **3–8 weeks**.
- Nervous system develops from – **ectoderm**.
- Placenta weight at term – **~500 g (⅙ of baby)**; cotyledons – **15–20**.
- Amniotic fluid at term – **~800–1000 ml**.
- Naegele's rule – **LMP + 9 months + 7 days**.
- Menopause hormone marker – **↑ FSH**.
- Lowest point in female peritoneal cavity – **pouch of Douglas**.

# 13 Quick Revision – Master Tables & 150 One-Liners

## 13.1 Largest, Smallest, Longest

| Record | Answer |
|---|---|
| Largest organ | Skin |
| Largest internal organ / gland | Liver |
| Largest endocrine gland | Thyroid |
| Largest lymphoid organ | Spleen |
| Largest bone / longest | Femur |
| Smallest bone | Stapes |
| Largest muscle | Gluteus maximus |
| Longest muscle | Sartorius |
| Smallest muscle | Stapedius |
| Strongest muscle | Masseter (uterus – strongest by force in labour) |
| Largest cell | Ovum |
| Smallest cell | Sperm (among smallest) |
| Longest cell | Neuron |
| Largest artery / vein | Aorta / inferior vena cava |
| Longest vein | Great saphenous |
| Largest nerve | Sciatic |
| Largest cranial nerve | Trigeminal (V) |
| Longest cranial nerve | Vagus (X) |
| Smallest cranial nerve | Trochlear (IV) |
| Largest WBC / most common WBC | Monocyte / neutrophil |
| Largest salivary gland | Parotid |
| Largest part of brain | Cerebrum |
| Largest joint | Knee |
| Hardest substance | Tooth enamel |
| Largest sesamoid bone | Patella |
| Largest lymph vessel | Thoracic duct |
| Largest paranasal sinus | Maxillary |
| Longest part of GI tract | Small intestine |
| Thickest skin / thinnest skin | Palms & soles / eyelids |

## 13.2 Normal Values – Master Table

| Parameter | Normal |
|---|---|
| **Body temperature** | **37 °C (98.6 °F)**; 36.5–37.5 °C |
| **Pulse** | **Adult 60–100**; newborn 120–160 |
| **Respiration** | **Adult 12–20**; newborn 30–60 |
| **BP** | **~120/80 mmHg**; MAP 70–105 |
| **$SpO_2$** | **95–100 %** |
| **Blood volume** | **5–6 L** (70–80 ml/kg) |
| **Hb** | **M 13–17, F 12–15 g/dl** |
| **RBC** | **M 4.5–5.5, F 4–5 million/µl** |
| **WBC** | **4,000–11,000/µl** |
| **Platelets** | **1.5–4.5 lakh/µl** |
| **ESR (Westergren)** | M 0–15, F 0–20 mm/hr |
| **Bleeding time / clotting time** | **2–7 min / 4–10 min** |
| **PT / INR** | **11–16 s / ~1** |
| **Fasting blood glucose** | **70–100 mg/dl** |
| **HbA1c** | **< 5.7 %** (diabetes ≥ 6.5 %) |
| **Serum $Na^+$ / $K^+$ / $Cl^-$** | **135–145 / 3.5–5.0 / 96–106 mEq/L** |
| **Serum calcium** | **8.5–10.5 mg/dl** |
| **Blood urea / creatinine** | **15–40 / 0.6–1.2 mg/dl** |
| **Serum bilirubin** | **0.3–1.2 mg/dl** |
| **Total protein / albumin** | **6–8 / 3.5–5 g/dl** |
| **Serum cholesterol** | **< 200 mg/dl** |
| **Uric acid** | **3–7 mg/dl** |
| **ABG pH / $PaCO_2$ / $HCO_3^-$ / $PaO_2$** | **7.35–7.45 / 35–45 / 22–26 / 80–100** |
| **GFR** | **125 ml/min** |
| **Urine output** | **1–1.5 L/day; ≥ 30 ml/hr** |
| **Urine sp. gravity / pH** | **1.010–1.025 / ~6** |
| **CSF volume / pressure** | **~150 ml / 60–180 mm$H_2O$** |
| **ICP** | **5–15 mmHg** |
| **IOP** | **10–21 mmHg** |
| **Cardiac output / SV** | **~5 L/min / ~70 ml** |
| **EF** | **55–70 %** |
| **Tidal volume / VC** | **500 ml / ~4.8 L** |
| **TBW** | **60 % (adult male)** |

| Parameter | Normal |
|---|---|
| CVP | 3–8 cm$H_2O$ (2–6 mmHg) |

## 13.3 Famous Names & Discoveries

| Person | Contribution |
|---|---|
| Andreas Vesalius | Father of Anatomy |
| Claude Bernard | Father of experimental physiology; milieu intérieur |
| William Harvey | Blood circulation (1628) |
| Walter Cannon | Homeostasis; fight or flight |
| Robert Hooke | Cell (1665) |
| Schleiden & Schwann | Cell theory |
| Watson & Crick | DNA double helix (1953) |
| Karl Landsteiner | ABO blood groups (1900–01); Rh (with Wiener, 1940) |
| Banting & Best | Insulin (1921) |
| Bayliss & Starling | Secretin – first hormone (1902) |
| Thomas Addison | Father of endocrinology; Addison's disease |
| Ivan Pavlov | Conditioned reflex (digestion) |
| Willem Einthoven | ECG |
| Hans Berger | EEG |
| Teasdale & Jennett | Glasgow Coma Scale (1974) |
| Huxley | Sliding filament theory |
| Frank & Starling | Frank-Starling law of heart |

## 13.4 150 Rapid-Fire One-Liners

**150 one-liners**

1. Structural & functional unit of life – **cell**.
2. Powerhouse of cell – **mitochondria**.
3. Suicide bags – **lysosomes**.
4. Protein synthesis – **ribosomes**.
5. Packaging – **Golgi apparatus**.
6. Cell membrane model – **fluid mosaic**.
7. Chromosomes – **46**; gametes – **23**.
8. Down syndrome – **trisomy 21**.
9. Mitosis – **2 diploid cells**; meiosis – **4 haploid**.
10. Plane – right & left – **sagittal**.
11. Plane – front & back – **coronal**.
12. Appendix – **right iliac fossa**.
13. Spleen – **left hypochondrium**.
14. Urinary bladder epithelium – **transitional**.
15. Trachea epithelium – **pseudostratified ciliated columnar**.
16. Fluid connective tissue – **blood**.
17. Total bones – **206**.
18. Longest bone – **femur**; smallest – **stapes**.
19. Commonest fractured – **clavicle**.
20. Bone without articulation – **hyoid**.
21. Movable skull bone – **mandible**.
22. Anterior fontanelle closes – **12–18 months**.
23. Posterior fontanelle closes – **6–8 weeks**.
24. Cervical vertebrae – **7**.
25. Atlas – **C1**; axis – **C2**.
26. True ribs – **7 pairs**; floating – **2 pairs**.
27. Bone-forming cell – **osteoblast**.
28. Ball & socket – **shoulder, hip**.
29. Pivot joint – **atlanto-axial**.
30. Saddle joint – **thumb CMC**.
31. Largest joint – **knee**.
32. Ligament – **bone to bone**; tendon – **muscle to bone**.
33. Strongest tendon – **Achilles**.
34. Involuntary striated muscle – **cardiac**.
35. Contractile unit – **sarcomere**.
36. Calcium binds – **troponin**.
37. NMJ transmitter – **acetylcholine**.
38. Rigor mortis – **no ATP**.
39. Main inspiratory muscle – **diaphragm**.
40. Infant IM site – **vastus lateralis**.

41. Safest IM site – **ventrogluteal**.
42. Pacemaker – **SA node**.
43. AV node rate – **40–60/min**.
44. Thickest wall – **left ventricle**.
45. Bicuspid valve – **mitral**.
46. Oxygenated vein – **pulmonary vein**.
47. Deoxygenated artery – **pulmonary artery**.
48. S1 – **AV valve closure**.
49. S2 – **semilunar valve closure**.
50. Cardiac cycle – **0.8 s**.
51. CO – **5 L/min**.
52. P wave – **atrial depolarisation**.
53. QRS – **ventricular depolarisation**.
54. T wave – **ventricular repolarisation**.
55. Coronary filling – **diastole**.
56. Apex beat – **5th ICS, MCL**.
57. Exchange vessels – **capillaries**.
58. Resistance vessels – **arterioles**.
59. Baroreceptors – **carotid sinus, aortic arch**.
60. Infant pulse for CPR – **brachial**.
61. Foramen ovale → **fossa ovalis**.
62. Ductus arteriosus → **ligamentum arteriosum**.
63. Ductus venosus → **ligamentum venosum**.
64. Umbilical vessels – **2 arteries, 1 vein**.
65. Plasma – **55 %**.
66. Albumin – **oncotic pressure**.
67. RBC life – **120 days**.
68. Erythropoietin – **kidney**.
69. Most common WBC – **neutrophil**.
70. Allergy WBC – **eosinophil**.
71. Heparin & histamine – **basophil**.
72. Platelet life – **7–10 days**.
73. Factor IV – **calcium**.
74. Haemophilia A – **factor VIII**.
75. Universal donor – **O −ve**.
76. Universal recipient – **AB +ve**.
77. ABO – **Landsteiner**.
78. Anti-D – **within 72 hrs**.
79. Antibody crossing placenta – **IgG**.
80. Colostrum antibody – **IgA**.
81. First antibody – **IgM**.
82. Allergy antibody – **IgE**.
83. T-cell maturation – **thymus**.

84. Largest lymphoid organ – **spleen**.
85. Vaccination – **artificial active immunity**.
86. Little's area – **epistaxis**.
87. Adam's apple – **thyroid cartilage**.
88. Only complete ring – **cricoid**.
89. Carina – **T4/T5**.
90. Foreign body – **right bronchus**.
91. Right lung lobes – **3**.
92. Surfactant – **type II pneumocytes**.
93. Tidal volume – **500 ml**.
94. Dead space – **150 ml**.
95. Respiratory centre – **medulla**.
96. $CO_2$ mostly carried as – **bicarbonate**.
97. Kussmaul breathing – **DKA**.
98. Cheyne-Stokes – **heart failure/ ↑ ICP**.
99. Deciduous teeth – **20**.
100. Ptyalin – **salivary amylase**.
101. Parotid duct – **Stensen's**.
102. HCl – **parietal cells**.
103. Intrinsic factor deficiency – **pernicious anaemia**.
104. Pepsinogen – **chief cells**.
105. Most absorption – **jejunum**.
106. B12 absorption – **terminal ileum**.
107. Iron absorption – **duodenum**.
108. Bile – **no enzyme, emulsifies fat**.
109. Gall bladder contraction – **CCK**.
110. First hormone – **secretin**.
111. Vitamin K synthesis – **gut bacteria**.
112. Kupffer cells – **liver macrophages**.
113. Urea formation – **liver**.
114. Nephron – **functional unit of kidney**.
115. GFR – **125 ml/min**.
116. Glucose reabsorption – **PCT**.
117. ADH acts on – **collecting duct**.
118. Renin – **JG cells**.
119. Female urethra – **4 cm**.
120. Micturition centre – **S2–S4 (pons)**.
121. Main ICF cation – **$K^+$**.
122. Vomiting – **metabolic alkalosis**.
123. ROME – **Respiratory Opposite, Metabolic Equal**.
124. Myelin CNS – **oligodendrocytes**.
125. BBB – **astrocytes**.
126. Resting potential – **−70 mV**.

**127.** Vital centres – **medulla**.
**128.** Balance – **cerebellum**.
**129.** Temperature centre – **hypothalamus**.
**130.** Broca's area – **motor speech (frontal)**.
**131.** CSF – **choroid plexus; 500 ml/day**.
**132.** Spinal cord ends – **L1–L2**.
**133.** Knee jerk – **L3–L4**.
**134.** Cranial nerves – **12 pairs**; spinal – **31**.
**135.** Bell's palsy – **CN VII**.
**136.** Wrist drop – **radial nerve**.
**137.** Sympathetic – **thoracolumbar**.
**138.** GCS coma – **≤ 8**.
**139.** Night vision – **rods (rhodopsin)**.
**140.** Sharpest vision – **fovea**.
**141.** Myopia lens – **concave**.
**142.** Aqueous drainage – **canal of Schlemm**.
**143.** Hearing organ – **organ of Corti**.
**144.** Largest organ – **skin**.
**145.** Vitamin D synthesis – **skin**.
**146.** Master gland – **pituitary**.
**147.** Acromegaly – **GH excess (adult)**.
**148.** Tetany after thyroidectomy – **hypoparathyroidism**.
**149.** Fertilisation site – **ampulla**.
**150.** Ovulation – **LH surge, 14 days before menses**.



**Exam strategy – Asha Didi ki last advice**

- GSSSB Staff Nurse paper me **Anatomy & Physiology se ~6–10 direct questions** aate hain, aur MSN/ OBG/ Paediatrics ke questions bhi isi base par hote hain.
- Sabse zyada pooche jaate hain – **largest/ smallest/ longest, normal values, heart & conduction, blood groups & immunity, cranial nerves, hormones & glands, kidney/ nephron, fetal circulation, menstrual cycle**.
- **Normal values ka master table** roz 5 min revise karo – yahi values clinical questions me bhi kaam aati hain.
- Diagrams (heart, nephron, brain, respiratory tree) ek baar khud draw karke dekho – yaad jaldi hota hai.
- MCQ Bank PDF ke saare questions **2 baar** solve karo aur galat hue questions ko notebook me likho.

— Anatomy & Physiology · GSSSB Staff Nurse Preparation · Padhte raho, revise karte raho! —